A Novel Spectral Index for Photovoltaic Array Mapping Using Satellite Imagery Across Complex Deployment Scenarios and Temporal Variations

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***Abstract*—The accurate spatiotemporal information on photovoltaic (PV) installations is essential for sustainable energy management and solar energy potential assessment. Satellite-based remote sensing provides an effective approach for large-scale and continuous monitoring of PV installations. However, due to variations in deployment scenarios and installation types, accurately detecting PV arrays from complex and heterogeneous backgrounds is still a major challenge. To address this issue, this study develops a novel spectral index method, the advanced photovoltaic index (APVI), based on Sentinel-2 satellite imagery. Through a systematic spectroscopic analysis of PV under six deployment scenarios and one snow-covered scenario, three key spectral characteristics of PV were identified at spatiotemporal scales: 1) high-reflectance difference between the two shortwave infrared bands (SWIR1 and SWIR2); 2) low-reflectance value in the near-infrared (NIR) band; and 3) a flat spectral shape from the red to NIR bands. The three key spectral features were used to construct APVI by mathematically combining four sensitive bands. Qualitative and quantitative evaluations across 16 cases in China demonstrate that APVI significantly outperforms five existing indices. It achieved an average F1 score of 95.2% across various deployment scenarios. In particular, APVI improves accuracy by 29.3%–34.6% over existing methods in snow-covered scenarios. In addition, APVI shows good threshold stability and cross-spatiotemporal adaptability. Overall, this study comprehensively considers the heterogeneity of PV arrays in medium-resolution imagery and provides a promising approach for the rapid and accurate detection of PV installations across complex scenarios at the satellite scale.**

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I. Introduction

IN ORDER to mitigate climate change and ensure energy security, the global community has set ambitious climate goals to transition from carbon-intensive fuels to renewable energy sources in a timely manner [1]. The development of photovoltaic (PV) energy is a key strategy for reducing carbon emissions and achieving carbon neutrality [2], [3], [4], [5]. In recent years, PV projects have experienced rapid growth. By 2023, the global installed PV capacity had reached 1411 gigawatts (GWs), accounting for approximately 25% of the total installed renewable energy capacity worldwide [6]. According to the International Energy Agency's Sustainable Development Scenario, global PV capacity is projected to reach 4240 GW by 2040 [7]. This implies that PV deployment will occupy an increasing amount of land resources. Given the continued expansion of PV installations, timely and accurate mapping of their spatiotemporal distribution has become increasingly important for effective deployment planning and sustainable land management.

Remote sensing technology provides a promising approach to obtaining the spatiotemporal distribution of PV installations. High-resolution imagery (typically 0.3-m per pixel or finer) from aerial or satellite platforms can be used to map PV installations with high accuracy and delineate detailed boundaries [8], [9]. For example, the aerial imagery can be used to map distributed PV installations deployed on rooftops [10], [11].Thanks to high-resolution RGB aerial imagery, small-scale PV arrays and individual PV panels on rooftops can be accurately identified [12], [13]. Airborne imaging spectroscopic data, with its rich spectral information, can further enhance the accuracy of PV detection [14]. For centralized PV systems, the boundaries of PV panels can also be accurately delineated using deep learning and high-resolution Google Earth imagery [15], [16]. However, such high-resolution aerial and satellite images are often difficult to obtain for free and have limited spatiotemporal coverage. Therefore, existing studies primarily utilize high-resolution data to extract PV installations at fine spatial scales (e.g., panel level) within target areas.

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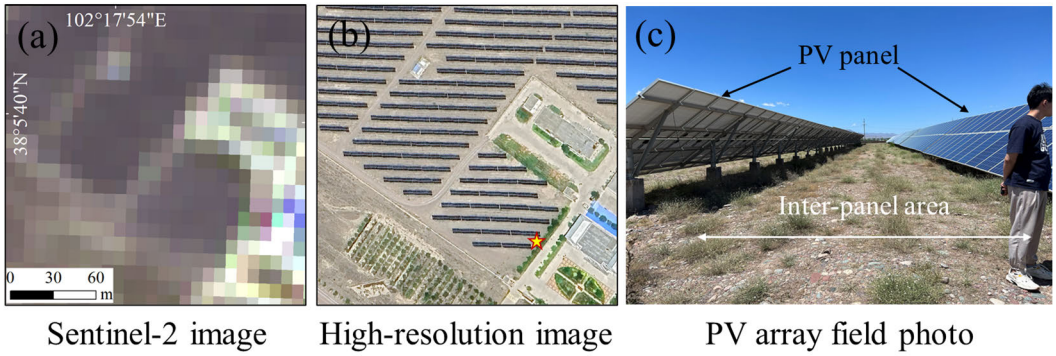


Fig. 1. (a) Sentinel-2 true color image of the PV array (R:G:B = B4:B3:B2), (b) high-resolution Google Earth imagery, and (c) field photograph.

Multispectral satellite images with medium resolution, such as Sentinel-2 (10 m) or Landsat (30 m), offer free access and global availability, making them ideal data sources for long-term continuous Earth observation [17]. The use of medium-resolution satellite images for large-scale spatiotemporal PV mapping has experienced notable advancements in recent years [18], [19], [20], [21], [22]. Due to the long-historical record of Landsat, researchers have used Landsat imagery and the random forest algorithm on the Google Earth Engine (GEE) platform to map the spatiotemporal distribution of PV installations at provincial or national scales [23], [24], [25], [26], [27]. In addition, some studies employ relatively high-resolution satellite imagery (e.g., Sentinel-2) together with deep learning or machine learning algorithms to extract PV arrays and then combine these results with historical imagery from other sources at coarser spatial resolutions to detect the timing of PV deployment [28], [29]. This approach balances the spatial accuracy of extraction and the temporal coverage of detection. The integration of multisource data (e.g., Sentinel-1 and Sentinel-2) with machine learning also presents an effective strategy for PV mapping [5], [30], [31], [32]. Although these studies have provided valuable methodologies and datasets for spatiotemporal PV extraction, classifier-based methods are computationally intensive and require large amounts of manually labeled training samples, which may pose challenges for timely data updates.

Spectral indices are an effective and efficient approach for target identification by exploiting distinctive spectral characteristics. Since more than 95% of global solar PV technology is based on the crystalline silicon (c-Si) [33], the optical properties of c-Si PV panels provide a representative basis for spectral-index construction. In particular, c-Si PV panels generally exhibit relatively low reflectance, an intrinsic design characteristic realized through the silicon absorber and antireflective coating to suppress reflection losses and enhance the solar energy absorption. As shown in Fig. S1, this low-reflectance behavior extends across much of the range of 400–900 nm, broadly consistent with the major terrestrial solar-irradiance range under standard AM1.5 conditions [34]. In addition, under pure spectral conditions, PV panels exhibit relatively higher blue-band reflectance within the visible region [35], [36]. Some studies have developed PV indices based on the blue characteristics of PV at the satellite scale, but these indices can only be used for specific deployment scenarios, such as PV on water surfaces [37], [38]. Other studies have constructed indices based on the pure spectral characteristics of PV panels and successfully applied them to satellite-scale imagery. For example, Stid et al. [36] developed the normalized difference PV index (NDPVI) and normalized blue deviation (NBD) based on the high-blue reflectance and low-NIR reflectance characteristics of PV panels in 0.6-m spatial resolution aerial imagery. These indices were successfully applied to Landsat imagery to identify PV and detect deployment times in the agricultural fields of California's Central Valley. He et al. [39] used a high-spectral-resolution ASD FieldSpec3 field spectroradiometer to collect and analyze spectral data of PV panels. They found a unique reflection peak in the range of 400–500 nm that can distinguish PV panels from most land backgrounds. Then, a universal solar PV panel index (SPPI) was developed by combining the strong absorption of PV panels in the visible to NIR (400–900 nm) range and the high reflectance characteristics in the shortwave infrared (1300–2500 nm). This index was applied to Sentinel-2 imagery, achieving efficient PV mapping.

Although these studies have effectively utilized the spectral characteristics of PV panels in pure conditions when constructing indices, they overlook the contribution and heterogeneity of land cover between PV panels in medium-resolution images. This may lead to some uncertainty when directly applying these indices to satellite images, especially in areas with larger spacing between PV panels. Given that the spatial resolution of medium-resolution images generally ranges from 10 to 30 m, satellites are unable to directly monitor the scale of individual PV panels and instead primarily monitor PV at the array scale. The pixels of PV arrays are composed of PV panels and interpanel areas (also known as the "underlying surface") (see Fig. 1). Interpanel land cover varies across PV deployment scenarios, affecting spectral characteristics and increasing the variability of PV spectra. In addition, the underlying surface land cover reduces the spectral difference between PV arrays and the surrounding background land types, which increases the difficulty of distinguishing them. Accurately extracting PV arrays from different deployment scenarios is more challenging. However, to the best

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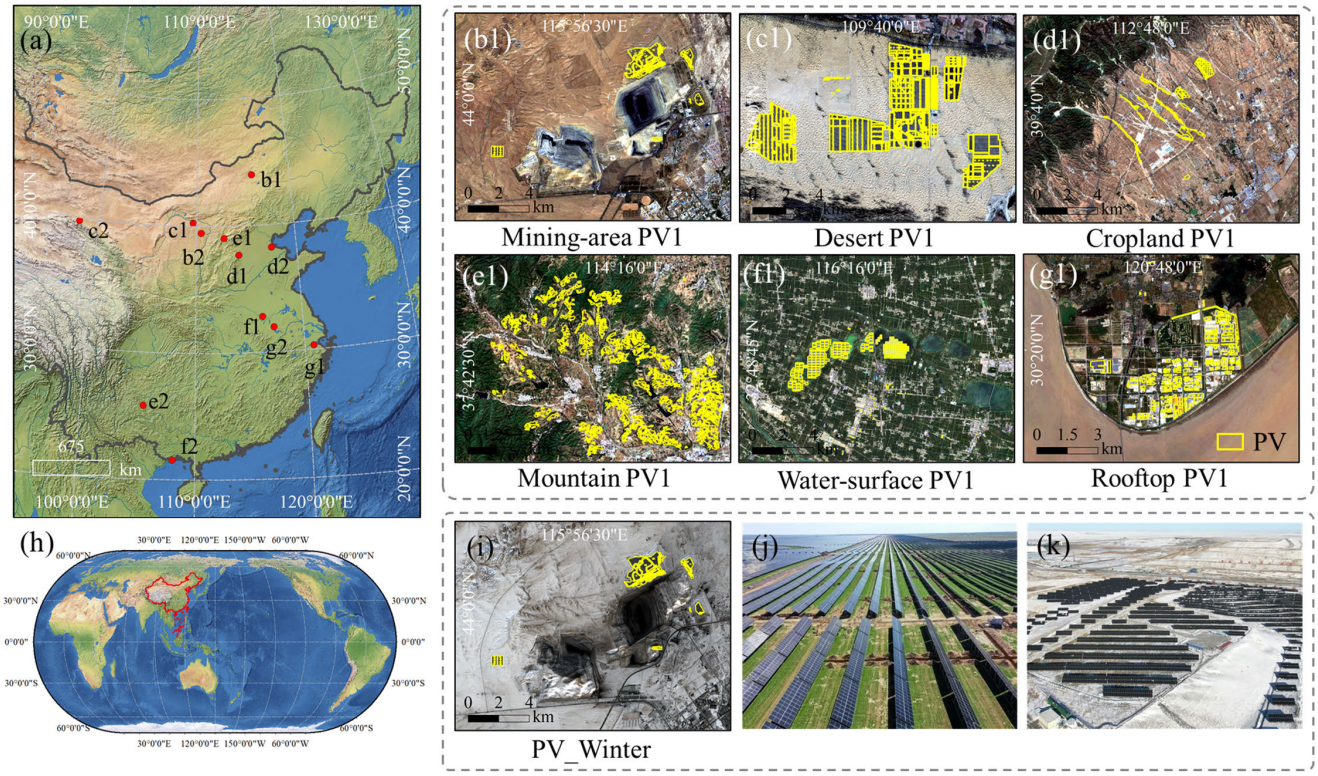


Fig. 2. Locations of the 12 study sites in China. (a) and (h) Distribution of study areas in China and worldwide, (b1) and (b2) mining-area PV, (c1) and (c2) DPV, (d1) and (d2) CPV, (e1) and (e2) MTPV, (f1) and (f2) WPV, and (g1) and (g2) rooftop PV. (i) Snow-covered mining-area PV in winter. (j) and (k) Ground photographs of PV arrays in nonwinter and winter conditions. Yellow outlines indicate PV array boundaries.

of our knowledge, there is a lack of systematic analysis on the heterogeneity of PV in different scenarios, and the spectral characteristics and change patterns of PV in medium-resolution satellite imagery remain unclear. In such a situation, developing a spatially robust PV index that accounts for the heterogeneity of the underlying surfaces of PV arrays is especially important.

In addition, c-Si, the material commonly used in PV panels, exhibits long-term stability and does not undergo significant changes over time [33]. However, the land cover of the underlying surface is highly dynamic and seasonal, which may cause PV pixels in satellite imagery to vary over time. For example, in winter at mid- to high-latitude regions, snow covers not only the PV panels but also the underlying surface, directly altering the spectral characteristics of PV. In addition, the lower solar elevation angle in winter increases the shadowed area on the underlying surface between PV rows, thereby enhancing the shadow contribution to the mixed spectral response of PV arrays [40], [41]. These factors further exacerbate the heterogeneity of PV pixels at the satellite scale. Indices developed based on PV panel characteristics may not effectively address the heterogeneity caused by these temporal variations. Therefore, this study also analyzes the spectral characteristics of PV under snow-covered scenarios to explore the temporal changes of PV at the satellite scale. This facilitates long-term continuous PV monitoring while expanding the temporal scope of monitoring.

To address the issue that existing indices overlook the contribution and heterogeneity of underlying surfaces, this study proposes a novel index. Sentinel-2 imagery from six typical PV deployment scenarios and one snow-covered scenario was selected to analyze the spectral heterogeneity of PV and to extract key spectral features for index construction (see Section III). The proposed index was then comprehensively compared with existing indices through visual assessment, separability analysis, and mapping accuracy evaluation to assess its performance across various scenarios (see Sections IV and V). Section VI further discusses threshold sensitivity, Landsat compatibility, and cross-national adaptability. This study aims to develop a robust approach for extracting PV arrays across complex scenarios.

II. Materials

A. Study Area

China is the country with the largest PV deployment globally, accounting for approximately one-third of the world's installed capacity [28], providing a rich basis for studying the diversity of PV deployment scenarios. In this study, six typical deployment scenarios within China are selected: mining areas, deserts, croplands, mountains, water surfaces, and urban rooftops. Two case locations are selected for each scenario to further capture the variations in PV deployment forms and background environments. For example, in the water-surface scenario, both floating water PV on inland lakes and stationary water PV on coastal aquaculture ponds are included. Fig. 2 presents the Sentinel-2 true-color images of the first case location (b1)–(g1) for each scenario, while Fig. S2 shows the corresponding second locations (b2)–(g2). Table S1 provides

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close-up images of all 12 case locations along with detailed descriptions of each scenario type.

Specifically, for mining-area PVs (MPVs), both the scale and spatial distribution of mining areas were considered. The first case site is located in Xilin Gol, Inner Mongolia, and features a large open-pit coal mine [see Fig. 2(b1)]. PV arrays are deployed on the dump site, with sparse vegetation planted between the panels to facilitate mine reclamation and ecological restoration. The second case site is in Ordos, Inner Mongolia, where PV arrays are distributed across dump sites and coal mining subsidence areas [see Fig. S2(b2)]. The surrounding area contains many small-open-pit coal mines.

For desert PVs (DPVs), the types of deserts were considered. The first case site is in the Kubuqi Desert in Ordos, Inner Mongolia [see Fig. 2(c1)]. As the seventh largest desert in China, it hosts the largest DPV power station in the country, which was built for sand fixation. The second case site is in Jiayuguan, Gansu, which features a Gobi-type (gravel) desert, in contrast to the sandy desert in Ordos [see Fig. S2(c2)].

For cropland PVs (CPVs), two scenarios of farmland cultivation and fallow were considered. The first case site is located in Xinzhou, Shanxi [see Fig. 2(d1)]. Due to the barren land, the PV arrays are installed in fallow farmland for rural poverty alleviation, with the background primarily consisting of bare soil. The second case site is in Cangzhou, Hebei, where the PV arrays are installed in cultivated farmland with crops planted [see Fig. S2(d2)]. The background here is primarily cultivated land with crops (e.g., wheat).

For mountain PVs (MTPVs), the vegetation coverage of mountainous areas was considered. The first case site is in Shijiazhuang, Hebei, where PV arrays are installed on slopes with lush vegetation [see Fig. 2(e1)]. The second case site is in Anshun, Guizhou, where PV arrays are deployed on severely rocky, desertified mountains for rocky desertification control [see Fig. S2(e2)]. The vegetation in this area is relatively sparse.

For water-surface PVs (WPVs), both the PV installation methods and the differences in the background water bodies were considered. The first case site is located in Fuyang, Anhui, where extensive coal mining subsidence areas have formed due to long-term underground coal extraction [see Fig. 2(f1)]. Floating water PV arrays are installed on idle water surfaces to support the aquavoltaics. The second case site is in Fangchenggang, Guangxi, where stationary water PV arrays are installed on coastal aquaculture ponds [see Fig. S2(f2)].

For rooftop PVs (RPVs), both the types of building roofs and the background differences were considered. Unlike the large-scale centralized PV deployments in other scenarios, rooftop PV belongs to distributed PV, with more scattered distribution and much smaller PV array sizes. The first case site is located in Haining, Zhejiang, where the "PV + commercial and industrial" model has been promoted [see Fig. 2(g1)]. PV arrays are deployed on the roofs of large factories with diverse roof colors. The second case site is in Hefei, Anhui, where PV arrays are installed on urban building rooftops [see Fig. S2(g2)]. This inland city has a high density of factories, and many buildings are equipped with blue color-coated steel tile roofs.

PV arrays in China are deployed across diverse land types, spanning a wide range of latitudes, and encompassing significant climatic and environmental differences. Due to these varying deployment contexts, PV arrays exhibit noticeable differences in color and texture characteristics. In Sentinel-2 true-color imagery, PV arrays can become nearly indistinguishable from background features in certain scenarios. The blue-band characteristic of PV panels is not obvious in Sentinel-2 imagery, which can be attributed to their relatively low blue reflectance compared with several background land-cover types and the influence of underlying surfaces on the observed PV spectral response.

Moreover, four selected case areas [see Fig. 2(a), (b1), (c1), (e1), and (e2)] are affected by snow accumulation during winter (see Fig. S3). As shown in Fig. 2(b1) and (i), even at the same location, the appearance of mining-area PV arrays differs markedly between snow-covered and nonwinter conditions. In snowy scenes, PV arrays appear dark black, while background land types are simplified into two main tones: black and white. Overall, the selected study areas encompass a wide variety of PV deployment settings, including snow-covered conditions.

B. Data

1) Sentinel-2 Data: The Sentinel-2 satellite, launched by European Space Agency (ESA), provides a valuable dataset for terrestrial mapping. In this study, the Sentinel-2 multispectral instrument (MSI) surface reflectance product (COPERNICUS/S2_SR_HARMONIZED) was obtained through the GEE platform. This dataset consists of images from two satellites, Sentinel-2A and Sentinel-2B, which provide 12 bands ranging from visible light to shortwave infrared. Detailed band information is shown in Table S2.

Given the significant expansion of PV deployment in recent years, the Sentinel-2 surface reflectance images for each case site were collected from 2023 or 2024. Specifically, the images for WPV1 and MPV2 are from 2024, while the remaining cases use imagery from 2023. A cloud cover threshold of 10% was applied. For the 12 cases across six deployment scenarios, median composite images from nonwinter months (March–November) were used to ensure temporal consistency and minimize the impact of rapid construction and outliers [30]. For the four snow-covered cases (MPV1_winter, DPV1_winter, CPV1_winter, and CPV2_winter), median composites from winter months (December 2023–February 2024) were used.

It should be noted that images from the first case site of each deployment scenarios (i.e., MPV1, DPV1, CPV1, MTPV1, WPV1, and RPV1), along with two snow-covered cases (i.e., MPV1_winter and CPV1_winter), were used to construct the sample spectral library, perform separability analysis, and design the PV index [see Sections III–V-B]. All 16 case images were used for visual evaluation and PV mapping accuracy assessment [see Section V-A and V-C].

2) Reference Data: Some existing global and national PV datasets are freely available. However, since these datasets were generated using different sensor types, acquisition dates, and spatial resolutions, directly using them as reference data may introduce uncertainties. Therefore, instead of adopting

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existing PV inventories, reference data were generated through manual visual interpretation. Assisted by high-resolution satellite imagery (e.g., Google Earth and World Imagery Wayback), and Sentinel-2 false-color composites (R:G:B = SWIR1:NIR:Red), PV vector layers were manually delineated. The resulting reference data served two main purposes: 1) providing PV references for visual evaluation and mapping accuracy assessment and 2) supporting the generation of sampling points for accuracy assessment. Although the PV reference boundaries were delineated using high-resolution imagery and are relatively accurate, mixed pixels in medium-resolution images may cause uncertainty near array edges. Therefore, after stratified random sampling based on the reference data, high-resolution imagery was used to further verify the class labels of samples near PV boundaries.

III. Methodology

The core idea of developing a PV index for PV detection is to exploit its unique spectral characteristics and construct an index through mathematical combinations of sensitive spectral bands. This index should yield high values in PV areas and low values in non-PV backgrounds, thereby enabling effective separation between PV and background land cover types using a simple threshold. Different from the design process of existing indices, this study further considers the heterogeneity of PV in satellite imagery and conducts detailed spectral analysis on PV in various scenarios.

To achieve this, three key questions need to be addressed.

1) What are the spectral characteristics of PV across different scenarios in medium-resolution satellite imagery?
2) What unique spectral features distinguish various PV categories from background land cover types?
3) How can these features be used to construct an index that effectively separates PV from the background?

In addition, a table of acronyms was added to further standardize the use of abbreviations and improve terminology consistency, as shown in Table S3.

A. Spectral Analysis of PV and Background Types

1) Spectral Analysis of PV in Different Scenarios: To address the first question, PV spectral samples were collected across various scenarios. Specifically, 500 random PV samples were selected from each of the six deployment scenarios (MPV1, DPV1, CPV1, MTPV1, WPV1, and RPV1), based on the PV reference data, to analyze PV spectral characteristics across different deployment scenarios. In addition, 500 samples were collected from each of two snow-covered cases (MPV1_winter and CPV1_winter), resulting in 1000 PV samples for analyzing PV spectral features under snow conditions.

As shown in Fig. 3, PVs in different scenarios exhibit clear spectral heterogeneity, particularly in the visible-to-NIR range. The spectra of DPV, WPV, RPV, and PV_winter show an almost flat trend (with little variation in reflectance) from the red band (B4) to the NIR band (B8). In contrast, other PV categories exhibit a higher reflectance in the NIR band compared to the visible bands. This difference is likely related to the land cover of the underlying surface. For PV deployed in mining areas and abandoned farmland (i.e., MPV and CPV), the underlying surfaces are mostly bare soil, including spoil dumps and bare cropland, resulting in more similar spectral characteristics. DPV exhibits higher reflectance values in the red-to-NIR range (B4~B8), which may be attributed to the strong reflectance characteristics of the underlying bare sand surface [42]. MTPV is more sensitive in the red-edge bands, with a greater spectral slope between the red and NIR bands, which may be related to the abundant vegetation on the underlying surface. The underlying surface of WPV includes floatation materials (high-density polyethylene) [43] and water, resulting in a relatively flat spectral shape from visible to NIR (B2~B8). The underlying surface of RPV consists of various materials (such as concrete, bitumen, and color-coated steel tiles), resulting in a similarly flat spectral shape from visible to NIR, but with higher reflectance values. For PV_winter, the PV exhibits high reflectance from visible to NIR (B2~B8), with mean reflectance values exceeding 0.22. This is likely due to the PV being covered by snow, which increases the reflectance.

Notably, in the shortwave infrared region (SWIR1~SWIR2), all PV types exhibit consistent spectral characteristics: reflectance in the SWIR1 band (B11) is higher than that in the SWIR2 band (B12). This suggests that the shortwave infrared bands, SWIR1 and SWIR2, are relatively insensitive to the underlying surface, making them useful spectral features for constructing indices to reduce the intraclass variation of PV.

2) Spectral Comparison of Different PV and Background Types: To address the second issue, spectral sampling was conducted for background land cover types. Seven background classes were selected from the eight case study images, namely: mine, bareland, impervious surface area (ISA), vegetation, water, winter-h, and winter-l. The mine class refers to open-pit mining areas and was sampled exclusively in the MPV1 study region, with a total of 500 samples. Winter-h and winter-l represent high-reflectance features (e.g., snow) and low-reflectance surfaces (e.g., shadows) in snowy winter scenes, respectively. They represent the main land features in snow-covered scenarios. These two classes were sampled exclusively in the MPV1_winter and CPV1_winter regions, with 1000 samples each. The remaining four classes (bareland, ISA, vegetation, and water) are common land cover types and were sampled across all nonsnow scenes. It is important to note that bareland includes various types of exposed soil, such as deserts, bare cropland, and bare surfaces in mountainous areas. ISA includes rooftops of different colors and roads in both urban and rural areas. Vegetation includes a wide range of plant-covered areas, such as grasslands, croplands, forests, and urban green spaces. Water covers diverse water bodies, including ponds, canals, lakes, rivers, and oceans. The sample sizes for these classes varied slightly according to their spatial proportions in different study areas: 1800 samples for bareland, 1000 for ISA, 1400 for vegetation, and 1400 for water.

As shown in Figs. 4 and S4, the reflectance spectral curves of various PV categories and background land cover types are presented. As expected, due to the contribution of the

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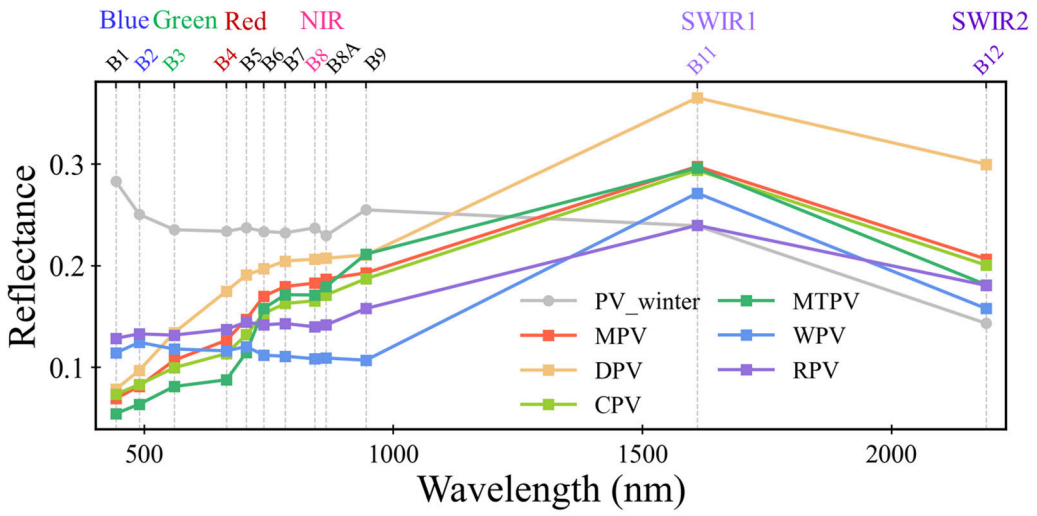


Fig. 3. Reflectance curves (mean values) of PVs in different scenarios. Note: B1 (coastal aerosol), B2 (blue), B3 (green), B4 (Red), B5 (red edge 1), B6 (red edge 2), B7 (red edge 3), B8 (NIR), B8A (red edge 4), B9 (water vapor), B11 (SWIR 1), and B12 (SWIR 2). Vertical dashed lines indicate the central wavelengths of Sentinel-2 bands.

underlying surface, PV categories exhibit certain spectral similarities with background land cover types. For example, in the visible to NIR range, MTPV displays characteristics like the "red edge" of vegetation, while WPV shows low reflectance values similar to those of water. Despite these similarities, PV categories still have unique spectral features that differentiate them from background land cover types. Due to the spectral absorption characteristics of the antireflective coating on PV panels, compared with land cover types other than mine, water, and winter-l, PV categories generally exhibit lower reflectance in the NIR band (B8). However, when compared with mine, water, and winter-l, all PV categories show a more pronounced reflectance difference in the shortwave infrared region (i.e., B11 and B12), with B11 reflectance significantly higher than B12. In contrast, these three background classes show relatively minor differences between the two bands.

Overall, the high reflectance difference in the shortwave infrared region (B11 and B12) is not only a common spectral signature across all PV categories but also a key feature for distinguishing PV categories from background land cover types. In addition, low reflectance value in the NIR band (B8) of PV categories is another distinguishing feature.

B. Separability Analysis and Index Construction

1) Jeffries–Matusita Distance (JMD): To address the third issue, it is necessary to further quantify the separability between PV categories and background land cover types. For this purpose, the JMD was employed, a statistical measure that assesses the dissimilarity or separability between two classes based on their probability distributions [44]. The JMD has been widely used in remote sensing as a standard criterion for evaluating class separability [45], [46], [47]. The JMD can be formally expressed as follows:

$$\mathrm{JM}_{ij} = \sqrt{2\left(1 - e^{-\alpha}\right)} \tag{1}$$

wherein

$$\alpha = \frac{1}{8}\left(\mu_i - \mu_j\right)^t \left\{\frac{C_i + C_j}{2}\right\}^{-1} \times \left(\mu_i - \mu_j\right) + \frac{1}{2}\ln\left\{\frac{\left|\left(C_i + C_j\right)/2\right|}{|C_i|^{1/2}\left|C_j\right|^{1/2}}\right\} \tag{2}$$

where i and j represent the two classes being compared, μ_i and C_i refer to the mean vector and covariance matrix of class i, respectively, and $|C_i|$ indicates the determinant of the covariance matrix C_i.

The value range of JM_{ij} is between 0 and $(2)^{1/2}$ [48]. For ease of analysis, its square is usually used to represent the range, namely, [0, 2] [48], [49]. A higher JMD value indicates better separability between classes [50]. Typically, when the JMD is greater than 1.8, it indicates good separability between classes, which also means that the within-class differences are smaller than the differences between classes. A JMD below 1.0 indicates poor separability between classes. When the JMD value is between 1.0 and 1.8, it indicates a certain degree of confusion between classes [51], [52].

2) Development of the Advanced Photovoltaic Index (APVI): The statistical distribution of PV categories and background land cover types across various spectral bands and band combinations is shown in Fig. 5. It can be observed that, in the B11 and B12 band combination, the values of the PV categories are generally higher than those of the background land cover types, indicating that B11 and B12 is an effective feature for constructing a PV index. However, relying solely on B11 and B12 may also increase the band combination values of vegetation and bare land, making it difficult to distinguish PV categories from these background types.

Specifically, as shown in Table I, in B11 and B12 band combination, the JMD values between the PV categories and vegetation are all below 1.0, and the JMD between RPV and bare land is only 0.509. To further reduce the confusion between PV and vegetation or bare land, another spectral

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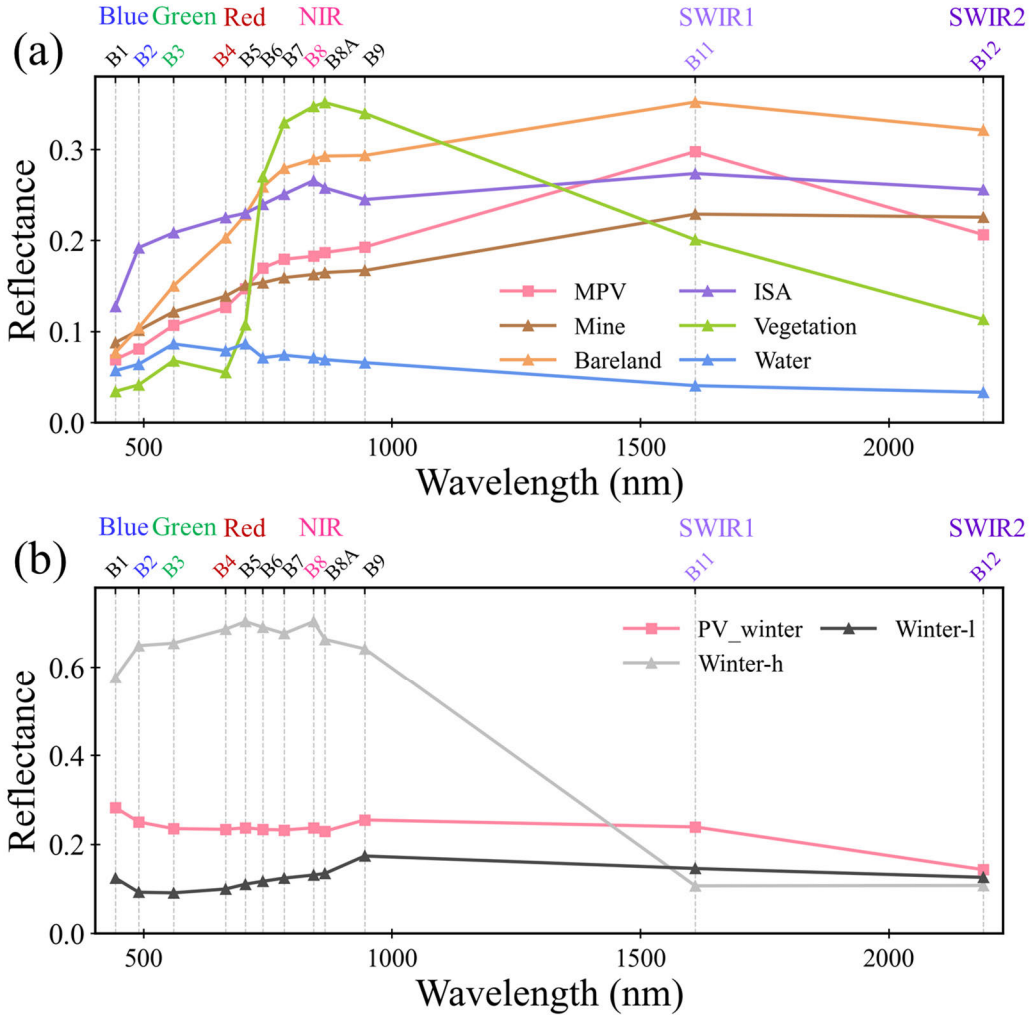


Fig. 4. Reflectance spectral comparison (mean values) of MPV, PV_winter, and representative background land-cover types. (a) MPV and background land-cover types, including mine, bareland, ISA, vegetation, and water. (b) PV_winter and winter background types, including winter high-reflectance background (Winter-h) and winter low-reflectance background (Winter-l). Vertical dashed lines indicate the central wavelengths of Sentinel-2 bands. Note: B1, coastal aerosol; B2, blue; B3, green; B4, red; B5, red edge 1; B6, red edge 2; B7, red edge 3; B8, NIR; B8A, red edge 4; B9, water vapor; B11, SWIR1; and B12, SWIR2.

feature of PV was introduced, namely, low reflectance in the NIR band (B8). In this band, all PV categories exhibit significantly lower reflectance than vegetation.

By constructing the band combination (B11 and B12)/B8, these two spectral features can be effectively integrated. As expected, the separability between PV categories and vegetation is significantly improved in (B11 and B12)/B8, as shown in Table I. The JMD values between the four PV categories (MPV, CPV, MTPV, and WPV) and vegetation are all close to 1.8, indicating good separability between categories. Meanwhile, the separability between PV categories and other background land cover types is also enhanced to some extent.

However, a good PV index should be able to highlight all PV categories while suppressing various background land cover types, in order to achieve effective discrimination between PV and background. According to the JMD analysis, the separability between the three PV categories and background land cover types is relatively low (JMD < 1.0) under the (B11 and B12)/B8 combination. Specifically, this includes between: 1) DPV and background types (vegetation and water); 2) RPV and vegetation; and 3) PV_winter and winter-l. Interestingly, these three PV categories (DPV, RPV, and PV_winter) exhibit a flat spectral shape from the red band (B4) to the NIR band (B8), as shown in Fig. 3. This characteristic can be quantified and normalized using the absolute value index abs[(B8 – B4)/(B8 + B4)]. As illustrated in Fig. 5, the index values of DPV, RPV, and PV_winter are close to 0, whereas those of the background land cover types (vegetation, water, and winter-l) are relatively high.

Since PV categories exhibit low values in abs[(B8 – B4)/(B8 + B4)] and high values in (B11 and B12)/B8, introducing abs[(B8 – B4)/(B8 + B4)] as the denominator helps integrate these two spectral features [see (3)]. This leads to higher index values for PV categories that possess both characteristics. The

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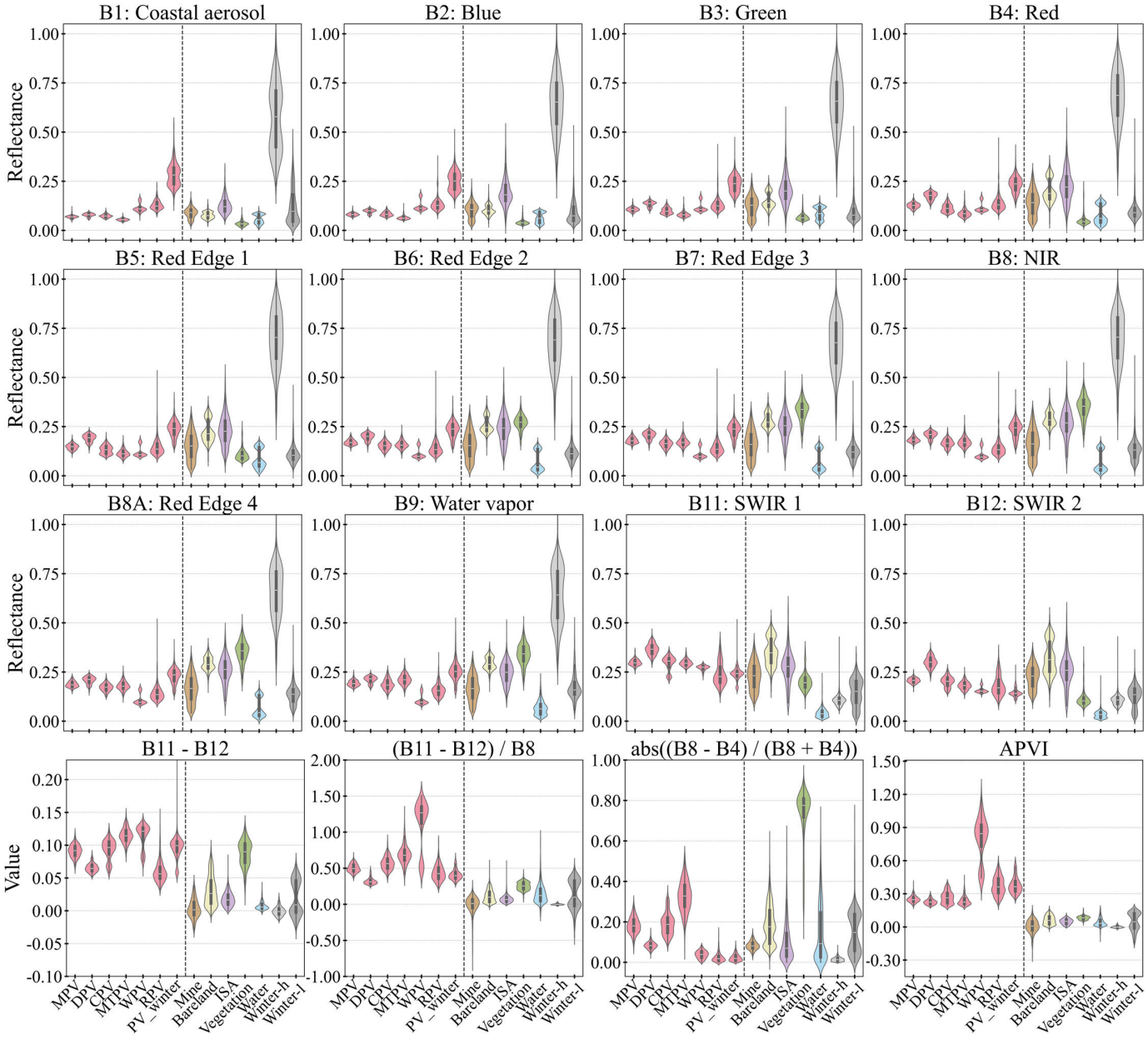


Fig. 5. Statistical distributions of PV categories and background land-cover types across Sentinel-2 spectral bands and derived band combinations. The distributions are color-coded as follows: PV categories (MPV, DPV, CPV, MTPV, WPV, RPV, and PV_winter) in red; mine in brown; bareland in yellow; ISA in purple; vegetation in green; water in blue; and winter background types (Winter-h and Winter-l) in light and dark gray, respectively. The vertical dashed line separates PV categories from background land-cover types.

new index is called the APVI

$$\mathrm{APVI} = \frac{\rho_{\mathrm{SWIR1}} - \rho_{\mathrm{SWIR2}}}{\rho_{\mathrm{NIR}} + \left| \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}} \right|} \tag{3}$$

where ρ_{Red}, ρ_{NIR}, ρ_{SWIR1}, and ρ_{SWIR2} refer to the reflectance of red band (B4), the near-infrared band (B8), the short-wave infrared 1 band (B11), and the short-wave infrared 2 band (B12), respectively.

As shown in Table I, after incorporating this feature into the denominator of (B11–B12)/B8, the JMD values between DPV and background land cover types (vegetation and water) all exceed 1.8. The separability between RPV and vegetation increases from 0.972 to 1.922, and the separability between PV_winter and winter-l increases from 0.958 to 1.797. These results confirm the effectiveness of the added feature. For other PV categories that do not exhibit a flat spectral shape between the red and NIR bands (i.e., MPV, CPV, and MTPV), their values in both B11 and B12 and B8 remain high. Therefore, even after introducing abs[(B8 – B4)/(B8 + B4)] into the denominator, they still maintain high separability from background land cover types.

Overall, the construction of APVI utilizes three key features: 1) the high-reflectance difference in the SWIR region (B11 and B12); 2) the low-reflectance value in the NIR band (B8); and 3) the flat spectral shape from the red to NIR bands (abs[(B8 – B4)/(B8 + B4)]). The proposed index achieves high

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TABLE I
JMD VALUE BETWEEN PV CATEGORIES AND BACKGROUND LAND COVER TYPES IN VARIOUS BAND COMBINATIONS AND SPECTRAL INDICES

Type	number	Band combinations and spectral indices			
		B11-B12	(B11-B12)/(B8+B12)	abs((B8-B4)/(B8+B4))	APVI
MPV	Mine	1.997	1.925	1.448	1.934
	Bareland	1.618	1.943	0.298	1.944
	ISA	1.997	1.999	0.551	1.998
	Vegetation	0.237	1.817	1.958	1.994
	Water	2.000	1.607	0.568	1.981
DPV	Bareland	1.132	1.409	1.116	1.908
	ISA	1.945	1.940	0.842	1.997
	Vegetation	0.822	0.385	1.995	1.990
	Water	2.000	0.917	1.066	1.967
CPV	Bareland	1.472	1.928	0.145	1.804
	ISA	1.967	1.995	0.427	1.940
	Vegetation	0.057	1.787	1.938	1.880
	Water	1.998	1.644	0.404	1.890
MTPV	Bareland	1.883	1.935	0.572	1.796
	ISA	2.000	1.988	1.128	1.960
	Vegetation	0.706	1.837	1.629	1.903
	Water	2.000	1.751	0.662	1.899
WPV	Bareland	1.740	1.840	1.317	1.957
	ISA	1.990	1.908	0.891	1.973
	Vegetation	0.430	1.785	1.998	1.967
	Water	1.999	1.740	1.107	1.965
RPV	Bareland	0.509	1.499	1.337	1.888
	ISA	1.407	1.842	0.853	1.951
	Vegetation	0.552	0.972	1.998	1.922
	Water	1.844	1.021	1.064	1.928
PV_wi	Winter-h	1.998	2.000	0.192	1.998
nter	Winter-l	1.561	0.958	1.208	1.797

separability between all PV categories and background land cover types.

IV. EXPERIMENTS

A. Comparison With Existing Indices

The proposed index was compared with five existing indices to evaluate its performance. These indices are: NDPVI (4) and NBD (5) proposed by Stid et al. [36]; the normalized difference built-up index (NDBI) (6) proposed by Zha et al. [53]; the SPPI (7) proposed by He et al. [39]; and the enhanced PV index (EPVI) (8) proposed by Wang et al. [54]. It should be noted that although NDBI is not an index specifically developed for PV, it is also often used for PV extraction [27], [30], [55], [56].

$$\mathrm{NDPVI} = \frac{0.5 \times \rho_{\mathrm{Blue}} - \rho_{\mathrm{NIR}}}{0.5 \times \rho_{\mathrm{Blue}} + \rho_{\mathrm{NIR}}} \tag{4}$$

$$\mathrm{NBD} = \frac{\rho_{\mathrm{Blue}} - \frac{\rho_{\mathrm{Red}} + \rho_{\mathrm{Green}}}{2}}{\rho_{\mathrm{Blue}} + \frac{\rho_{\mathrm{Red}} + \rho_{\mathrm{Green}}}{2}} \tag{5}$$

$$\mathrm{NDBI} = \frac{\rho_{\mathrm{SWIR1}} - \rho_{\mathrm{NIR}}}{\rho_{\mathrm{SWIR1}} + \rho_{\mathrm{NIR}}} \tag{6}$$

$$\mathrm{SPPI} = \frac{\rho_{\mathrm{Coastal}}}{\rho_{\mathrm{Green}}} \times \frac{(\rho_{\mathrm{SWIR1}} - \rho_{\mathrm{NIR}} - |\rho_{\mathrm{NIR}} - \rho_{\mathrm{Green}}|)}{(\rho_{\mathrm{SWIR1}} + \rho_{\mathrm{NIR}})} \tag{7}$$

$$\mathrm{EPVI} = \frac{2 \times \rho_{\mathrm{SWIR1}} - \rho_{\mathrm{SWIR2}} - \rho_{\mathrm{NIR}}}{2 \times \rho_{\mathrm{SWIR1}} + \rho_{\mathrm{SWIR2}} + \rho_{\mathrm{NIR}}} \tag{8}$$

where $\rho_{\mathrm{Coastal}}, \rho_{\mathrm{Blue}}, \rho_{\mathrm{Green}}, \rho_{\mathrm{Red}}, \rho_{\mathrm{NIR}}, \rho_{\mathrm{SWIR1}}$, and ρ_{SWIR2} refer to the reflectance of the coastal aerosol band (B1), blue band (B2), green band (B3), red band (B4), near-infrared band (B8), short-wave infrared 1 band (B11), and short-wave infrared 2 band (B12), respectively.

B. Evaluation of PV Mapping Results

To validate the PV mapping results, this study performed stratified random sampling for all 16 study cases based on the PV reference data described in Section II-B, aiming to reduce arbitrariness and bias in the accuracy assessment. For each case, 5000 sample points were randomly selected. Given that PV areas account for a relatively small proportion in most study regions, a compromise strategy was adopted [57], [58] to appropriately increase the number of samples from the rare class, ensuring that each case included at least 500 PV samples. Ultimately, a sample pool consisting of 8960 PV samples and 71 040 non-PV samples was established (see Fig. S5).

For each case study, PV mapping was carried out using a threshold-based segmentation method applied to the six indices. Specifically, pixels with values greater than the threshold were classified as PV, while those with lower values were classified as non-PV. The optimal threshold for each index was determined through an iterative approach: 10% of the total sample pool was first randomly selected as the test set; then, with a step size of 1/100 of the index value range, the F1 score was iteratively calculated for different thresholds. The threshold yielding the highest F1 score was selected as the final segmentation threshold. The remaining 90% of the samples were subsequently used to evaluate the accuracy of the mapping results [52], [59]. The quantitative evaluation of PV mapping results was conducted using the confusion matrix and its derived accuracy metrics, including user's accuracy (UA), producer's accuracy (PA), overall accuracy (OA), and the F1 score, as defined in the following equations:

$$\mathrm{UA} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{9}$$

$$\mathrm{PA} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{10}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP} + \mathrm{TN}} \tag{11}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{UA} \times \mathrm{PA}}{\mathrm{UA} + \mathrm{PA}} \tag{12}$$

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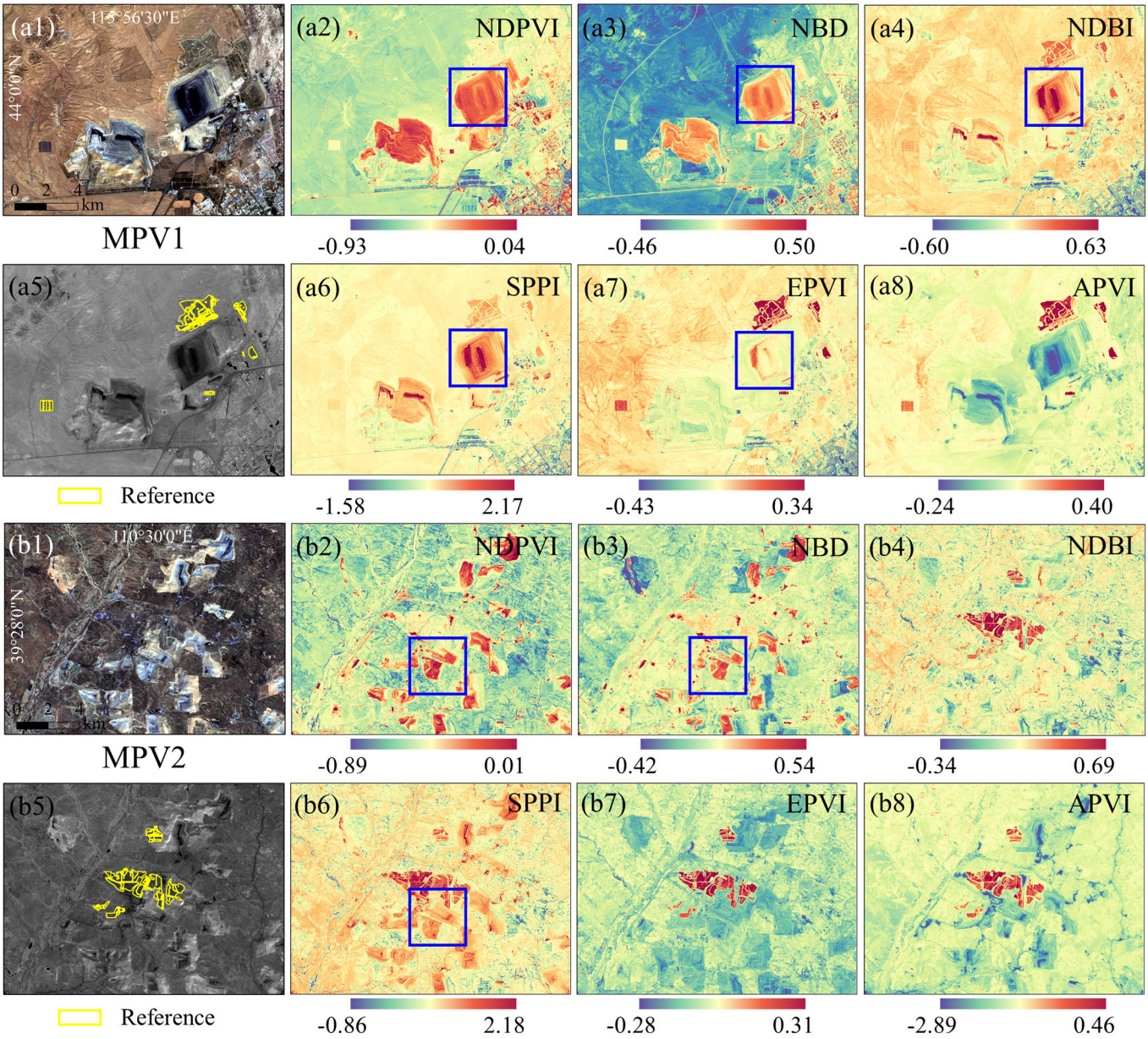


Fig. 6. Visual comparison of different indices for PV deployment in mining areas. (a1) and (b1) Sentinel-2 true-color images. (a5) and (b5) Reference PV boundaries. The index maps generated by (a2) and (b2) NDPVI, (a3) and (b3) NBD, (a4) and (b4) NDBI, (a6) and (b6) SPPI, (a7) and (b7) EPVI, and (a8) and (b8) APVI. Yellow outlines indicate the reference boundaries of PV arrays. Blue solid rectangles highlight representative non-PV areas that are falsely enhanced by existing indices.

where true positives (TPs), false negatives (FNs), false positives (FPs), and true negatives (TNs) refer to the number of PV pixels correctly detected, PV pixels omitted, non-PV pixels wrongly identified as PV, and non-PV pixels correctly excluded, respectively, based on the comparison between classification results and reference points.

V. Results

A. Visual Evaluation Results

A comprehensive comparison between the proposed index and five existing indices was conducted across all 16 study cases to evaluate the performance of APVI. Visual assessment, as an intuitive approach, clearly demonstrates the effectiveness of each index in enhancing PV pixels while suppressing background interference. In this study, a gradient color scheme was used to visualize the index images, where red indicates high values associated with PV and blue represents low values related to the background. In addition, since PV is difficult to visually identify in Sentinel-2 true color imagery, reference data were incorporated to facilitate a clearer comparative analysis.

1) Visual Assessment of Indices in the MPV Scenario (See Fig. 6): For PV arrays deployed in mining areas, the primary background land cover type that tends to be confused with PV is mining land. As shown in Fig. 6, both in the large open-pit coal mine case (MPV1) and in the coal mine cluster case (MPV2), NDPVI, NBD, and SPPI exhibit higher sensitivity to

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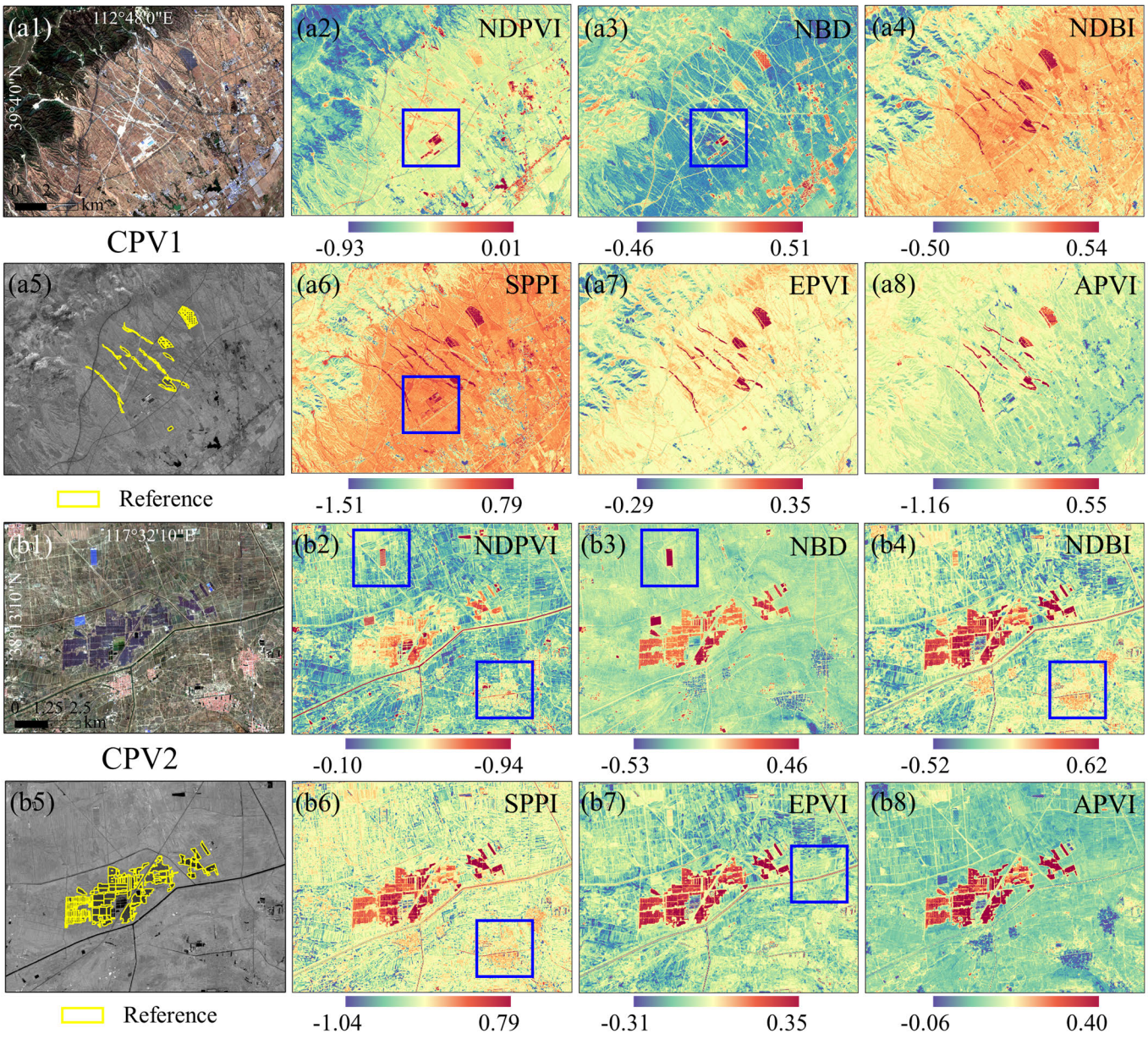


Fig. 7. Visual comparison of different indices for PV deployment in cropland areas. (a1) and (b1) Sentinel-2 true-color images. (a5) and (b5) Reference PV boundaries. The index maps generated by (a2) and (b2) NDPVI, (a3) and (b3) NBD, (a4) and (b4) NDBI, (a6) and (b6) SPPI, (a7) and (b7) EPVI, and (a8) and (b8) APVI. Yellow outlines indicate the reference boundaries of PV arrays. Blue solid rectangles highlight representative non-PV areas that are falsely enhanced by existing indices.

mining land, resulting in over-enhancement of non-PV mining areas in their corresponding index images, as highlighted by the blue solid rectangles. As clearly observed in the local detail views, the index values of mining land are even higher than those of PV (see Fig. S6). Notably, NDBI and EPVI perform poorly in MPV1 but relatively better in MPV2, possibly due to their sensitivity to terrain shadows caused by deep pits in large-scale open-pit mines. In contrast, APVI images highlight PV areas in red, making them clearly distinguishable from the background. This suggests that APVI not only enhances PV features but also effectively suppresses interference from background land cover types, particularly mining land.

2) Visual Assessment of Indices in the DPV Scenario (See Fig. S7): For PV arrays deployed in sandy desert areas (DPV1), all indices except NDPVI and NBD effectively highlight PV regions, with results largely consistent with the reference data. This may be attributed to the relatively simple and homogeneous desert background, facilitating the distinction between PV and the surroundings. In contrast, for the deployment in the Gobi (gravel desert) region (DPV2), only EPVI and APVI demonstrate clear advantages, while the other indices perform poorly. Specifically, NDPVI, NBD, NDBI, and SPPI exhibit higher sensitivity to industrial buildings near the Gobi and fail to suppress them effectively [see Figs. S7(b) andS8(a)]. Moreover, these indices (NDPVI, NBD,

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and NDBI) do not adequately enhance certain PV regions, resulting in significant confusion between PV arrays and the Gobi surface [see Figs. S7(b) and S8(b)]. This may be due to the subtle differences in the desert type of underlying surface. The underlying surface between the PV panels in Ordos mainly consists of a sandy desert composed of homogeneous fine sand grains. In contrast, the underlying surface in Jiayuguan is primarily a Gobi desert composed of gravel of varying sizes [60]. Existing studies have shown that gravel exhibits stronger absorption characteristics than sand, with lower reflectance across the visible to shortwave infrared spectrum [42]. This lower reflectance may cause PV arrays to exhibit spectral characteristics that are more similar to those of the Gobi background, making them more difficult to distinguish. It is worth noting that although both EPVI and APVI effectively suppress desert and Gobi backgrounds, APVI appears to have a stronger suppression capability. In the APVI images, large areas are rendered in dark green [see Fig. S7(a8) and (b8)], indicating a greater distinction between PV and background.

3) Visual Assessment of Indices in the CPV Scenario (See Fig. 7): For PV arrays deployed in cropland areas, whether on abandoned cropland (CPV1) or cultivated cropland (CPV2), the spatial distribution of different indices reveals broad consistency while also highlighting key differences. All indices perform well in suppressing most background land cover types, with the main differences lying in their ability to suppress impervious surfaces and waterbody edges. As shown in Fig. 7(a2), (a3), and (a6), NDPVI, NBD, and SPPI are more sensitive to impervious surfaces. Specifically, the local detail images show that many village houses surrounding farmland and field roads are erroneously highlighted in the same red color as PV arrays [see Fig. S9(a)]. In CPV2, NDPVI and NBD fail to effectively distinguish blue-roofed buildings from PV arrays [see Fig. 7(b2) and (b3)], while NDPVI, NDBI, and SPPI show greater sensitivity to red-roofed village building clusters[see Fig. 7(b2), (b4), and (b6)]. Narrow rivers and the edges of waterbodies are prone to generating mixed pixels, which interfere with target identification. As shown in Fig. S9(b), for drainage ditches around farmland, all indices except NBD and APVI fail to suppress this type of interference.

4) Visual Assessment of Indices in the MTPV Scenario (See Fig. S10): For PV arrays deployed in mountainous areas, the installations are more scattered and exhibit irregular shapes. In the case of PV deployed in vegetation-covered mountains (MTPV1), only the visualization results of EPVI and APVI show a spatial distribution of PV that closely matches the reference map. In contrast, the visualizations from the other indices exhibit significant confusion between PV and the complex background, making PV arrays difficult to identify visually [see Fig. S10(a)]. Specifically, these indices fail to effectively suppress fragmented bare land and mountainous infrastructure such as buildings and roads, which appear in colors similar to PV [see Fig. S11(a)]. For PV deployed in rocky desertified mountainous areas (MTPV2), where the background land cover is relatively simple, all indices except NDPVI and NBD clearly reveal the spatial distribution of PV [see Fig. S10(b)]. However, in terms of background suppression, EPVI and APVI outperform NDBI and SPPI [see Fig. S11(b)].

5) Visual Assessment of Indices in the WPV Scenario (See Fig. 8): For PV arrays deployed on water surfaces, the environmental conditions of the two study cases differ significantly. As shown in Fig. 8, APVI demonstrates a clear advantage, almost perfectly suppressing all background land cover types to deep blue in both cases, whereas other indices show notable limitations. Specifically, for floating water PV installed on water bodies formed by ground subsidence due to underground coal mining (WPV1), the surrounding water is distributed in the form of lakes, and the PV arrays are arranged in square formations. The PV arrays appear in varying shades of blue in the image, which may be related to the installation method and arrangement density of the floating PV systems. The different contributions of the floating box materials between the PV panels result in variations in pixel brightness. NDPVI, NBD, NDBI, and EPVI confuse these brighter PV arrays with water bodies [see Figs. 8(a) and S12(a)], indicating that intraclass differences exceed interclass differences for these indices, which may lead to the omission of some PV arrays during mapping. For stationary water PV deployed over coastal aquaculture ponds (WPV2), the PV appears in deep blue against the water background. NDPVI, NBD, and NDBI erroneously enhance the index values of certain sea areas [see Fig. 8(b)], while EPVI fails to effectively suppress mixed pixels at the pond edges [see Figs. 8(b) and S12(b)], potentially resulting in misclassification of some background areas as PV during mapping.

6) Visual Assessment of Indices in the RPV Scenario (See Fig. S13): For PV arrays deployed on rooftops, the complex coloration of rooftops is a major factor interfering with PV identification. Different colors indicate the diversity of roofing materials. In the RPV1 study area, PV arrays are installed on rooftops of various colors, such as red, white, and blue, among others. NDPVI and NBD exhibit stronger responses to white and blue rooftops than to PV, resulting in significant enhancement of non-PV rooftops [see Fig. S14(a)]. Although NDBI, SPPI, EPVI, and APVI all effectively highlight PV pixels, APVI performs better in suppressing background features [see Fig.S13(a)]. In the RPV2 study area, the urban environment is more complex, and PV arrays are more sparsely distributed. In the visualizations of NDPVI, NBD, NDBI, and SPPI, many background land cover types (such as urban water bodies and non-PV rooftop buildings) are incorrectly enhanced, leading to significant confusion between PV and background [see Fig. S13 (b)]. As a result, the spatial distribution of PV is almost unrecognizable in the index visualizations. Only EPVI and APVI produce results that are relatively consistent with the reference map. In terms of detail, APVI demonstrates stronger background suppression, whereas EPVI exhibits more noise in the imagery [see Figs. S13(b) and S14(b)].

7) Visual Assessment of Indices in the Snow-Covered PV Scenario (See Fig. 9): In the context of snow-covered PV scenarios, APVI demonstrates outstanding performance, effectively distinguishing PV from the background in all four study cases. As shown in Fig. 9, most of the background in the study

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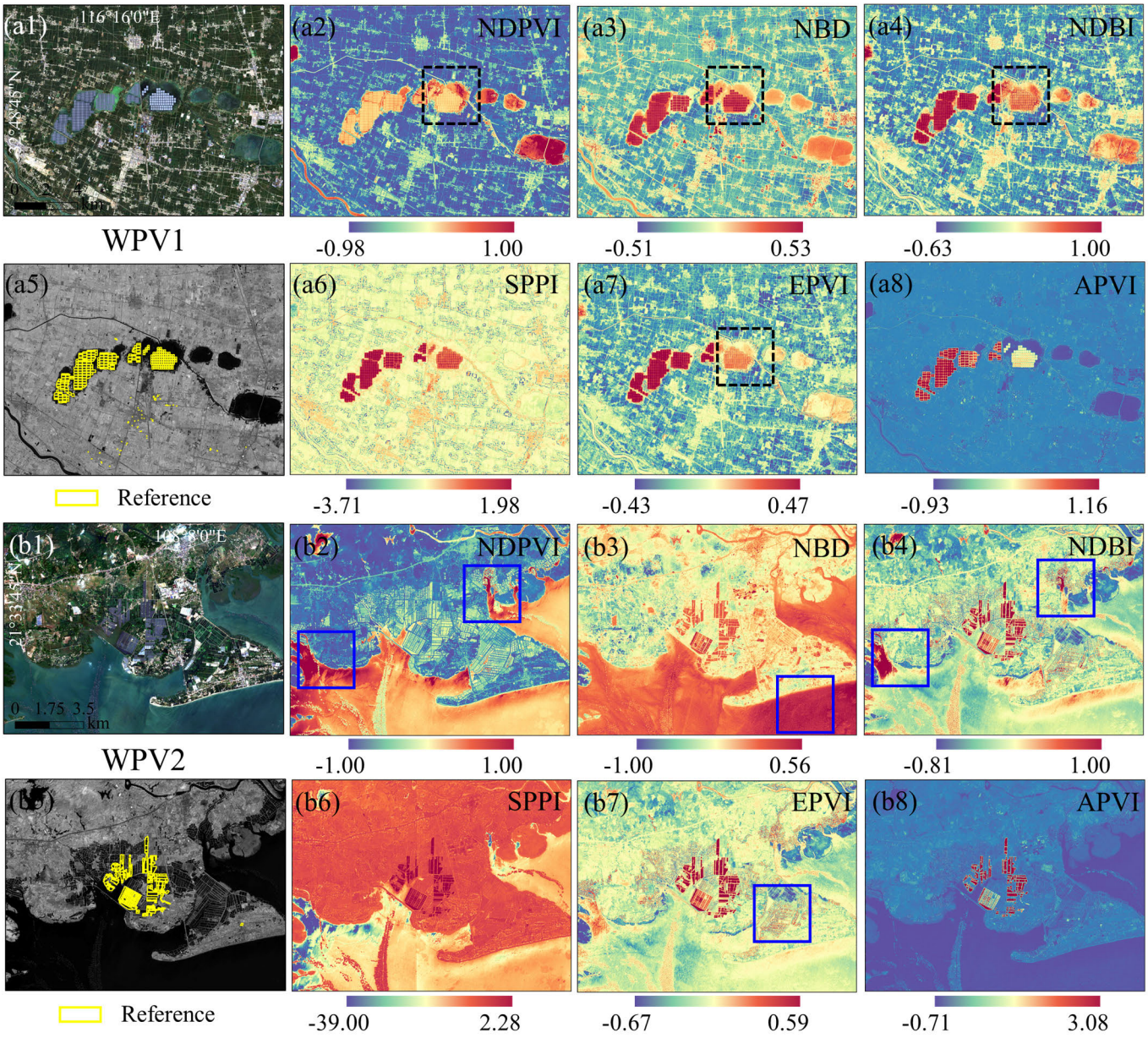


Fig. 8. Visual comparison of different indices for PV deployment on water surfaces. (a1) and (b1) Sentinel-2 true-color images. (a5) and (b5) Reference PV boundaries. The index maps generated by (a2) and (b2) NDPVI, (a3) and (b3) NBD, (a4) and (b4) NDBI, (a6) and (b6) SPPI, (a7) and (b7) EPVI, and (a8) and (b8) APVI. Yellow outlines indicate the reference boundaries of PV arrays. Blue solid rectangles highlight representative non-PV areas that are falsely enhanced by existing indices. Black dashed rectangles highlight representative PV areas that are insufficiently enhanced by existing indices.

area is covered by snow. In the visualizations of each index, PV and snow are clearly distinguishable. However, when it comes to handling dark surface features, the other five indices perform poorly compared to APVI. Specifically, background land cover types such as open-pit coal mines [see Fig. 9(a)], mountainous areas [see Fig. S15(a)], and built-up areas [see Fig. S15(b)] are significantly affected by topographic shadows, which are easily confused with PV and result in an intense red appearance in the visualizations of these indices. Moreover, when PV panels are heavily covered by snow, the PV signal is weakened, making it difficult for most indices to effectively highlight these PV arrays. For instance, in Fig. 9(b), the PV arrays on the right side of the image are heavily snow-covered and appear similar in color to the surrounding desert in the visualizations of most indices. Therefore, APVI not only exhibits strong shadow resistance but also has the unique capability to enhance weak PV signals under snow cover.

Overall, APVI exhibits the strong enhancement capabilities for PV arrays across various deployment scenarios and effectively suppresses diverse and complex background types. It is also the only index that can effectively identify PV arrays under snow-covered conditions. This suggests that APVI, developed with consideration of underlying surface contributions, is better suited to handle uncertainties caused by scene and temporal changes. In contrast, existing indices may enhance certain background features along with PV pixels,

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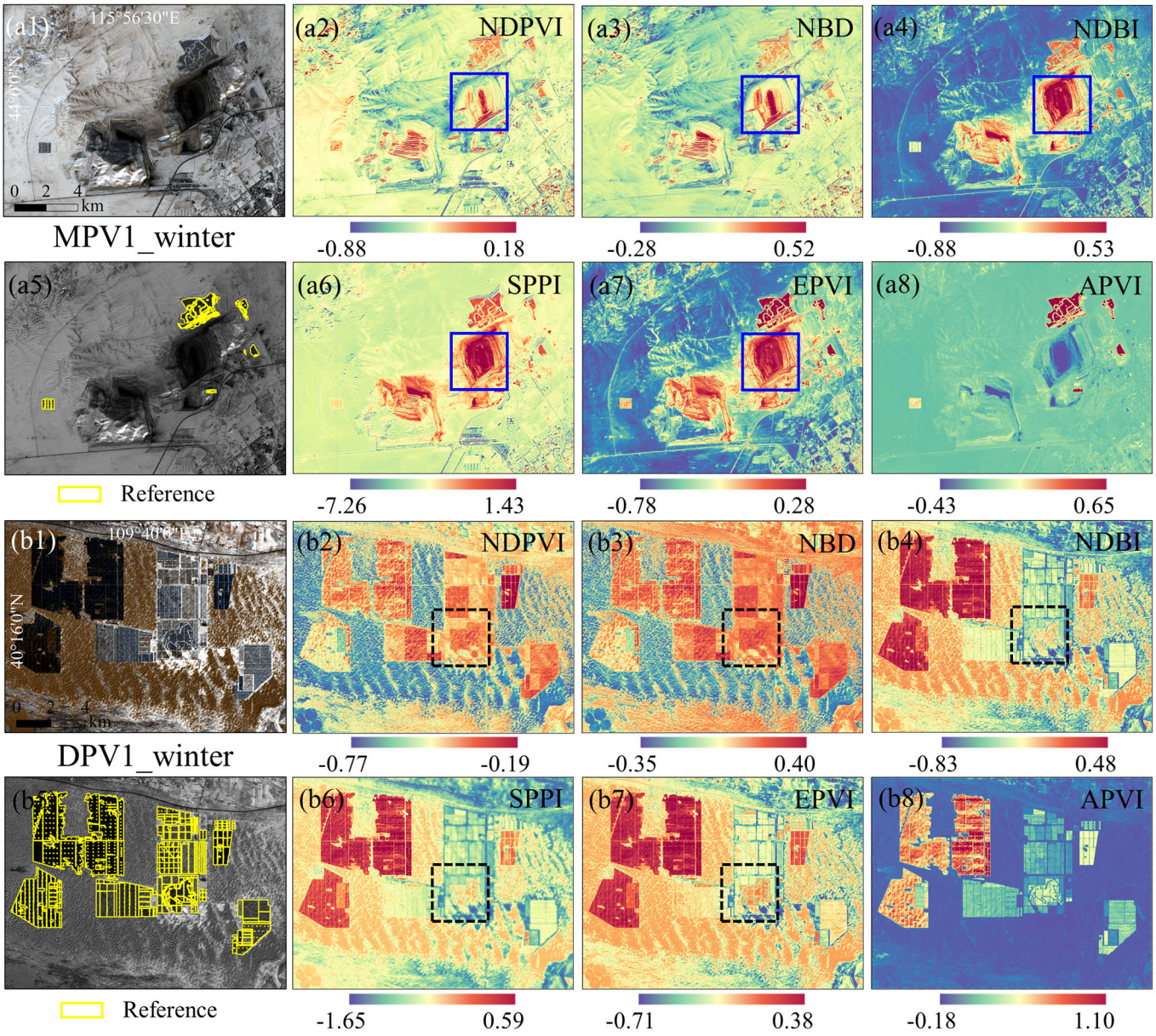


Fig. 9. Visual comparison of different indices in snow-covered PV scenarios. (a1) and (b1) Sentinel-2 true-color images. (a5) and (b5) Reference PV boundaries. The index maps generated by (a2) and (b2) NDPVI, (a3) and (b3) NBD, (a4) and (b4) NDBI, (a6) and (b6) SPPI, (a7) and (b7) EPVI, and (a8) and (b8) APVI. Yellow outlines indicate the reference boundaries of PV arrays. Blue solid rectangles highlight representative non-PV areas that are falsely enhanced by existing indices. Black dashed rectangles highlight representative PV areas that are insufficiently enhanced by existing indices.

thereby reducing their effectiveness in specific deployment scenarios.

B. Separability Analysis Results

Based on the spectral library constructed in Section III-A, a quantitative analysis was further conducted to assess the separability between PV and background types under different indices. Fig. 10 shows the statistical distributions of six indices across various PV categories and land cover types. Table S4 lists the JMD between PV and each background type under different indices.

Consistent with the visual assessment results, EPVI and APVI generally exhibit higher separability between PV and background types compared to the other four indices. Specifically, across the seven scenarios, NDPVI consistently performs well only in distinguishing PV from vegetation, with JMD values always exceeding 1.8. This is because the index was originally designed for PV extraction in agricultural areas, making it highly effective in separating vegetation but less effective for other land cover types. For NBD, the value distributions of PV and background types are largely overlapping, which limits its applicability to only certain scenarios. For example, only in the WPV scenario does the JMD between PV and the background exceed 1.0. The core feature used by these two indices is the high blue reflectance of the PV panels, which is interfered with by the underlying surface in medium-resolution satellite imagery (see Figs. 4 and S4). Therefore,

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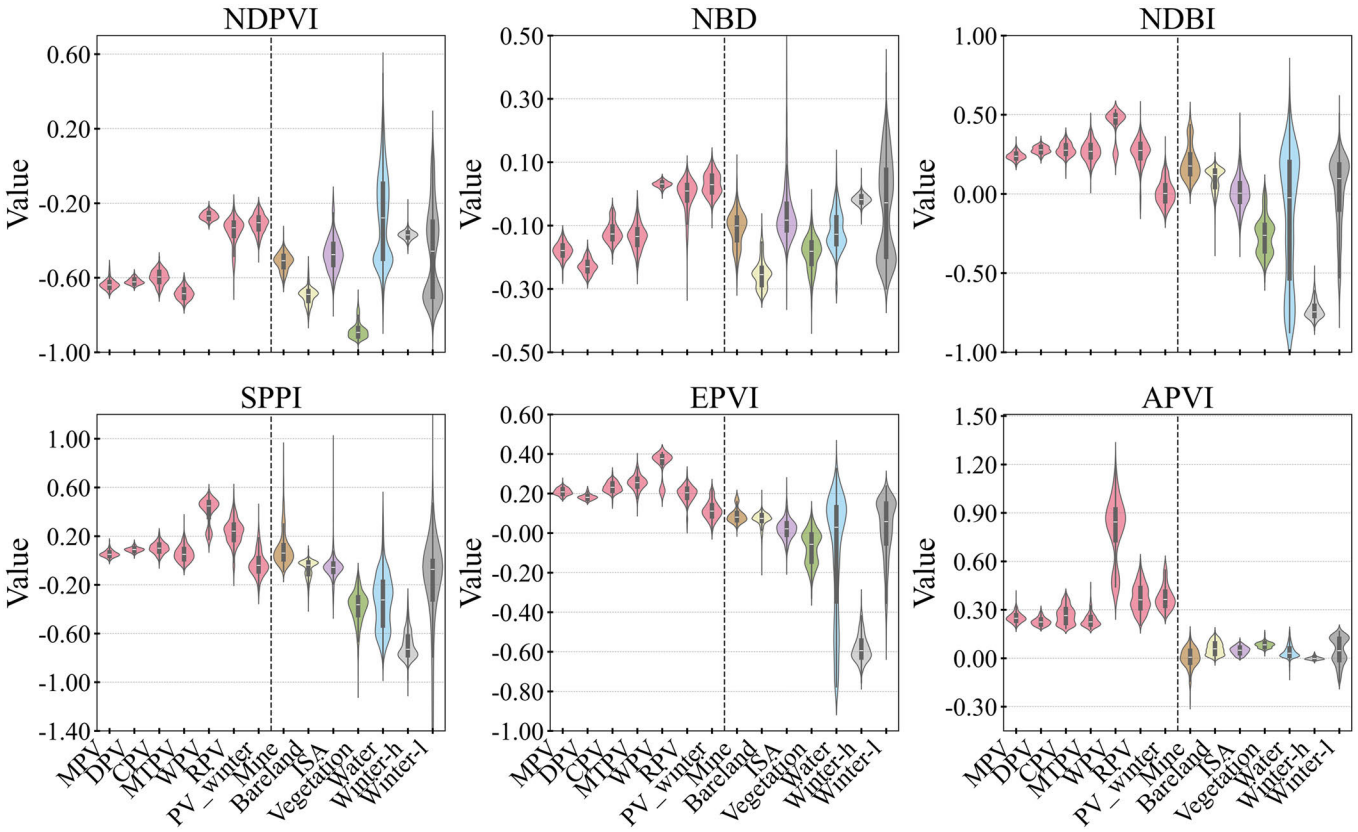


Fig. 10. Statistical distributions of PV and background types across different indices. The distributions are color-coded as follows: PV categories (MPV, DPV, CPV, MTPV, WPV, RPV, and PV_winter) in red, mine in brown, bareland in yellow, ISA in purple, vegetation in green, water in blue, and winter background types (Winter-h and Winter-l) in light and dark gray, respectively. The vertical dashed line separates PV categories from background land-cover types.

they are unable to effectively separate PV and the background in complex scenes.

In contrast, due to the use of PV features from the NIR to SWIR bands (e.g., B11–B8), NDBI and SPPI reduce the variability within the PV category, leading to more concentrated value distributions. This helps maintain a relatively stable separability in complex scenarios. However, their performance is still poor when dealing with certain background types (e.g., mine and ISA), with JMD values falling below 1.0. Overall, EPVI and APVI demonstrate outstanding performance in distinguishing PV from background types. In the six deployment scenarios, their JMD values are mostly above 1.5. In the snow-covered scenario, all indices except APVI show JMD values below 1.0 between PV and Winter-l. This is because the PV features used by existing indices (e.g., B11–B8 used by EPVI and NDBI) change in snow-covered scenes, making the indices unsuitable for this scenario.

C. PV Mapping Results

Based on the sample pool and methodology described in Section IV-B, optimal threshold selection and accuracy assessment were conducted for all 16 cases. Figs. S16 and S17 illustrate the F1 score curves and the corresponding optimal thresholds of each index under 12 deployment scenario cases and four snow-covered scenario cases, respectively. Using these optimal thresholds, index images were segmented to produce PV mapping results. The mapping accuracies for the two types of scenarios are presented in Tables II and S5. Overall, APVI achieved high mapping accuracy and low standard deviation across all 16 study cases. In the 12 deployment scenario cases, APVI significantly outperformed the other indices in PV mapping accuracy. Specifically, the average F1 scores and standard deviations of the six indices are as follows: NDPVI: 0.551 $\pm$ 0.199, NBD: 0.578 $\pm$ 0.214, NDBI: 0.828 $\pm$ 0.091, SPPI: 0.843 $\pm$ 0.104, EPVI: 0.891 $\pm$ 0.064, and APVI: 0.952 $\pm$ 0.020. Among them, APVI achieved higher average F1 scores than all other indices and improved by 6.1 percentage points over the best-performing existing index, EPVI, while maintaining the lowest standard deviation (0.020), demonstrating superior accuracy and stability.

Based on the mapping results, the performance of each index under different PV deployment scenarios was further analyzed. As shown in Fig. 11, in the MPV scenario, all indices except EPVI and APVI tend to misclassify mining land and infrastructure as PV (UA $<$ 0.70), whereas the PV mapping results produced by EPVI and APVI closely match the reference data. However, APVI exhibits less noise around the mining areas [see Fig. 11(b7)]. In the DPV scenario, all indices achieve relatively high mapping accuracy in the DPV1 case. However, noticeable differences can be seen in the Gobi PV case (DPV2). As illustrated in Fig. S18, NDPVI, NBD, NDBI, and SPPI tend to misclassify industrial buildings as PV,

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TABLE II
PV MAPPING ACCURACY OF EACH INDEX UNDER DIFFERENT DEPLOYMENT SCENARIOS

	Indices	Accuracy					Accuracy			
		UA	PA	OA	F1score		UA	PA	OA	F1score
MPV1	*NDPVI*	0.272	0.924	0.745	0.420	*MPV2*	0.190	0.787	0.643	0.306
	NBD	0.274	0.918	0.748	0.422		0.416	0.807	0.868	0.549
	NDBI	0.656	0.829	0.939	0.732		0.793	0.980	0.972	0.877
	SPPI	0.520	0.827	0.906	0.638		0.604	0.953	0.933	0.740
	EPVI	0.870	0.940	0.980	0.904		0.947	0.991	0.994	0.969
	APVI	0.927	0.982	0.990	0.954		0.959	0.991	0.995	0.975
DPV1	*NDPVI*	0.811	0.975	0.961	0.885	*DPV2*	0.406	0.840	0.861	0.548
	NBD	0.525	0.941	0.860	0.674		0.436	0.702	0.880	0.538
	NDBI	0.911	0.837	0.962	0.872		0.523	0.802	0.907	0.633
	SPPI	0.912	0.958	0.979	0.935		0.581	0.944	0.926	0.720
	EPVI	0.921	0.922	0.976	0.921		0.814	0.956	0.974	0.879
	APVI	0.947	0.958	0.985	0.953		0.925	0.987	0.991	0.955
CPV1	*NDPVI*	0.519	0.949	0.907	0.671	*CPV2*	0.555	0.791	0.916	0.653
	NBD	0.428	0.869	0.871	0.573		0.796	0.996	0.974	0.885
	NDBI	0.897	0.982	0.987	0.937		0.840	0.960	0.978	0.896
	SPPI	0.857	0.962	0.980	0.907		0.852	0.922	0.976	0.886
	EPVI	0.961	0.927	0.989	0.943		0.871	0.991	0.984	0.927
	APVI	0.954	0.976	0.993	0.965		0.876	0.991	0.985	0.930
MTPV1	*NDPVI*	0.416	0.709	0.871	0.524	*MTPV2*	0.649	0.873	0.940	0.744
	NBD	0.380	0.727	0.854	0.499		0.788	0.842	0.962	0.814
	NDBI	0.852	0.893	0.974	0.872		0.853	0.978	0.981	0.911
	SPPI	0.769	0.784	0.955	0.777		0.932	0.918	0.985	0.925
	EPVI	0.904	0.942	0.984	0.923		0.923	0.987	0.990	0.954
	APVI	0.916	0.922	0.984	0.919		0.973	0.960	0.993	0.966
WPV1	*NDPVI*	0.634	0.991	0.942	0.773	*WPV2*	0.184	1.000	0.557	0.311
	NBD	0.852	0.998	0.982	0.919		0.244	0.987	0.693	0.391
	NDBI	0.859	0.662	0.955	0.748		0.812	0.978	0.975	0.887
	SPPI	0.884	0.984	0.986	0.932		0.963	0.973	0.994	0.968
	EPVI	0.910	0.678	0.961	0.777		0.978	0.809	0.979	0.886
	APVI	0.928	0.998	0.992	0.961		0.972	0.989	0.996	0.980
RPV1	*NDPVI*	0.162	0.998	0.485	0.279	*RPV2*	0.372	0.749	0.848	0.497
	NBD	0.105	0.996	0.148	0.189		0.351	0.778	0.834	0.484
	NDBI	0.880	0.731	0.963	0.799		0.707	0.858	0.950	0.775
	SPPI	0.811	0.962	0.974	0.880		0.834	0.782	0.963	0.807
	EPVI	0.894	0.693	0.961	0.781		0.814	0.849	0.966	0.831
	APVI	0.927	0.953	0.988	0.940		0.909	0.938	0.984	0.923

and NDPVI, NBD, and NDBI also exhibit a certain degree of PV omission. For the CPV scenario (see Fig. S19), NDPVI, NBD, NDBI, and SPPI misclassify a large amount of villages as PV, while the remaining indices perform well overall. In the MTPV scenario, the mapping accuracy in vegetation-covered mountainous areas is slightly lower compared with that in rocky desertified mountainous areas. As shown in Fig. S20, the fragmented bare land, buildings, and roads in

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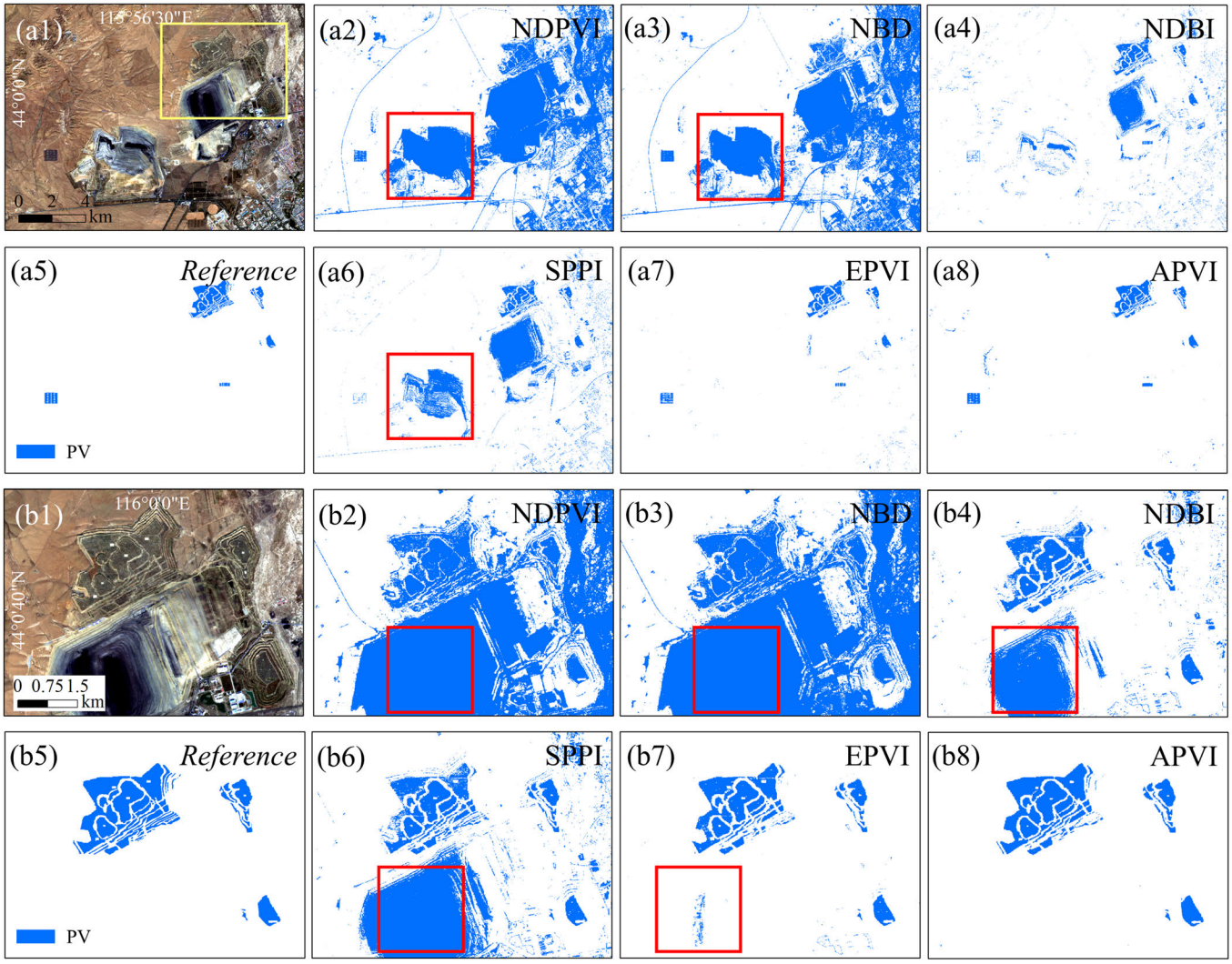


Fig. 11. PV mapping results of each index in the MPV1 case. (a1) and (b1) show the Sentinel-2 true-color images. (a5) and (b5) Reference PV maps. The remaining panels present the PV mapping results generated by different indices. The yellow rectangle in (a1) indicates the enlarged area shown in row (b). Red solid rectangles highlight falsely extracted regions. (a2) and (b2) NDPVI, (a3) and (b3) NBD, (a4) and (b4) NDBI, (a6) and (b6) SPPI, (a7) and (b7) EPVI, and (a8) and (b8) APVI.

these regions are the primary causes of misclassification for certain indices.

For the WPV scenario, there are significant differences in mapping performance among the indices (see Fig. 12). Specifically, NDPVI misclassifies many coal mining subsidence area water bodies and rivers as floating water PV (UA = 0.634), while NDBI and EPVI tend to omit parts of the PV (PA < 0.680). In contrast, NBD, SPPI, and APVI perform relatively well, all achieving F1 scores above 0.919. In the rooftop PV scenario, NDPVI and NBD exhibit extremely low mapping accuracy. In particular, for the RPV1 case, both indices yield F1 scores below 0.3. As shown in Fig. S21, the diverse roof colors in urban areas make it difficult for these two indices to effectively identify PV arrays.

In the four snow-covered scenario cases, the average F1 scores and their standard deviations for the six indices are as follows: NDPVI: 0.684 ± 0.048, NBD: 0.644 ± 0.091, NDBI: 0.630 ± 0.120, SPPI: 0.673 ± 0.091, EPVI: 0.689 ± 0.127, and APVI: 0.977 ± 0.013. Compared to the existing indices, APVI improves the F1 score by 29.3–34.6 percentage points. The smaller standard deviation further indicates that APVI exhibits stronger cross-regional robustness. Overall, APVI is the only index that maintains excellent PV mapping performance under snow conditions, while the other indices show significant limitations in these scenarios.

In snow-covered scenarios, two main factors influence the accuracy of PV mapping: 1) misclassification caused by terrain shadows and 2) omission of PV areas due to snow-covered panels. As shown in Fig. 13, all indices except APVI misclassify large amounts of mountain shadows as PV areas. This may be due to the self-shadowing effect of PV panels in winter, which strengthens the contribution of underlying surface shadows to PV pixels, thereby reducing the distinction between PV and terrain shadows. As shown in Fig. S22, DPV arrays, which were accurately identified in nonwinter seasons, are significantly missed when covered by snow in winter. This is because winter scenes not only introduce new sources of interference (such as stronger shadows) but also

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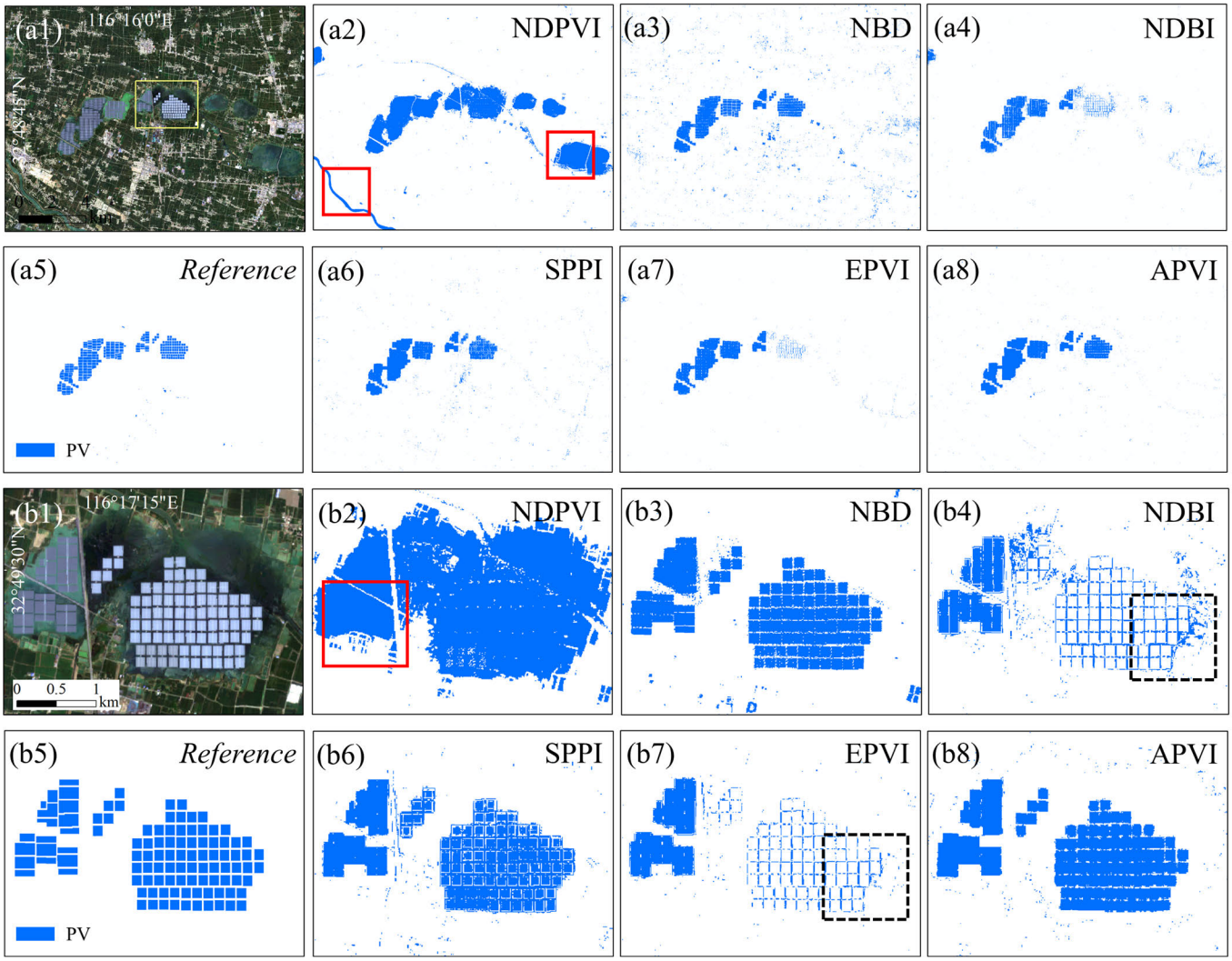


Fig. 12. PV mapping results of each index in the WPV1 case. (a1) and (b1) Sentinel-2 true-color images and (a5) and (b5) reference PV maps. The remaining panels present the PV mapping results generated by different indices. The yellow rectangle in (a1) indicates the enlarged area shown in row (b). Red solid rectangles highlight falsely extracted regions, while black dashed rectangles indicate omitted PV regions. (a2) and (b2) NDPVI, (a3) and (b3) NBD, (a4) and (b4) NDBI, (a6) and (b6) SPPI, (a7) and (b7) EPVI, and (a8) and (b8) APVI.

significantly alter the spectral characteristics of the target object (i.e., PV). The proposed APVI effectively overcomes these two key interference factors, demonstrating robust PV identification capabilities in snowy conditions.

VI. Discussion

A. Threshold Sensitivity Analysis

A robust index for detecting and quantifying PV should maintain relatively stable performance under moderate threshold variations across different deployment scenarios and observation conditions. To evaluate this, a threshold sensitivity analysis was conducted separately for nonwinter and winter snow-covered scenarios. For the nonwinter cases, the optimal thresholds from the 12 cases shown in Fig. S16 were averaged to obtain a mean threshold of 0.178. A series of thresholds within $\pm 50\%$ of this mean value was then applied to re-segment the data, and the results were evaluated using the same validation samples. As shown in Fig. 14, APVI exhibits different levels of threshold sensitivity across deployment scenarios. Overall, APVI maintains an F1 score of approximately 70% or higher when the threshold varies from -20% to $+10\%$ relative to the nonwinter mean threshold, corresponding to 0.142–0.196. This indicates that APVI can provide acceptable mapping performance within a moderate threshold interval in nonwinter deployment scenarios, although local threshold fine-tuning remains beneficial for obtaining optimal results.

For the winter snow-covered scenarios, the threshold sensitivity analysis was performed separately because the spectral responses of PV arrays and background land covers differ from those under nonwinter conditions. The average optimal threshold of APVI in the winter cases was 0.139. As shown in Fig. S23, when the threshold varies from -10% to $+50\%$ relative to this winter mean threshold, corresponding to 0.125–0.209, the minimum F1 score among all winter cases remains 75.2%. This result does not suggest that a broader threshold range improves mapping accuracy. Rather, it indicates that APVI maintains acceptable and relatively stable

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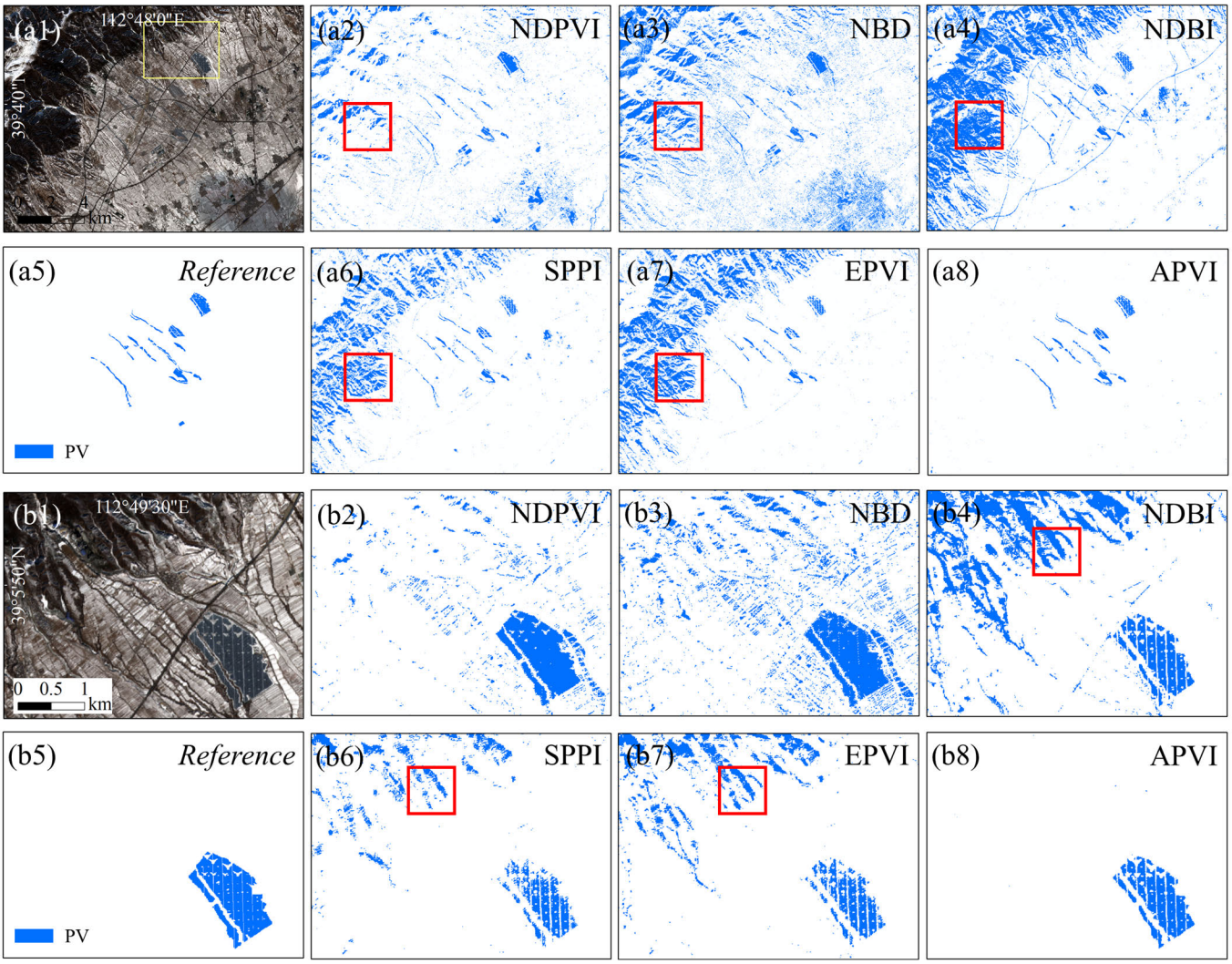


Fig. 13. PV mapping results of each index in the CPV1_winter case. (a1) and (b1) Sentinel-2 true-color images and (a5) and (b5) reference PV maps. The remaining panels present the PV mapping results generated by different indices. The yellow rectangle in (a1) indicates the enlarged area shown in row (b). Red solid rectangles highlight falsely extracted regions. (a2) and (b2) NDPVI, (a3) and (b3) NBD, (a4) and (b4) NDBI, (a6) and (b6) SPPI, (a7) and (b7) EPVI, and (a8) and (b8) APVI.

performance over a winter-specific threshold interval, which is shifted toward higher threshold values compared with the nonwinter scenarios.

B. Applicability of APVI Across Different Satellite Imagery and Seasons

Since APVI is constructed using spectral bands commonly available in medium-resolution satellites, it is expected to be applicable to other satellite imagery as well. In this study, Landsat 8 multispectral imagery with corresponding spectral bands was selected for further analysis, and the band correspondence between Sentinel-2 and Landsat 8 is provided in Table S6. This experiment was designed to evaluate the cross-sensor applicability of APVI rather than to conduct a strict pixel-by-pixel comparison between Sentinel-2 and Landsat 8. For Sentinel-2, APVI was calculated using B4, B8, B11, and B12 at a 10-m working resolution after resampling in GEE. For Landsat 8, APVI was calculated using the corresponding OLI bands B4, B5–B7 at the native 30-m spatial resolution. APVI was calculated from seasonal median-composited Landsat 8 imagery generated in GEE [61], [62], [63]. Specifically, all valid Landsat 8 observations within a given seasonal period were stacked, and the median reflectance value was calculated for each pixel and each band. This procedure reduces the influence of residual clouds, cloud shadows, sensor noise, and other extreme values, thereby providing a more robust seasonal representation of surface reflectance.

The MPV1 region was selected as a case study. In this area, PV arrays are deployed on spoil dumps, which are typically covered with vegetation for land reclamation, and exhibit pronounced phenological variations. In addition, the area is affected by snow cover during winter, making it a suitable example for illustrating seasonal changes in the underlying surface of PV arrays. The Sentinel-2 and Landsat 8 imagery used in this analysis covers the period from March 2023 to February 2024, with seasonal divisions defined as spring (March–May), summer (June–August), autumn (September–November), and winter (December–February). As shown in Fig. S24, APVI

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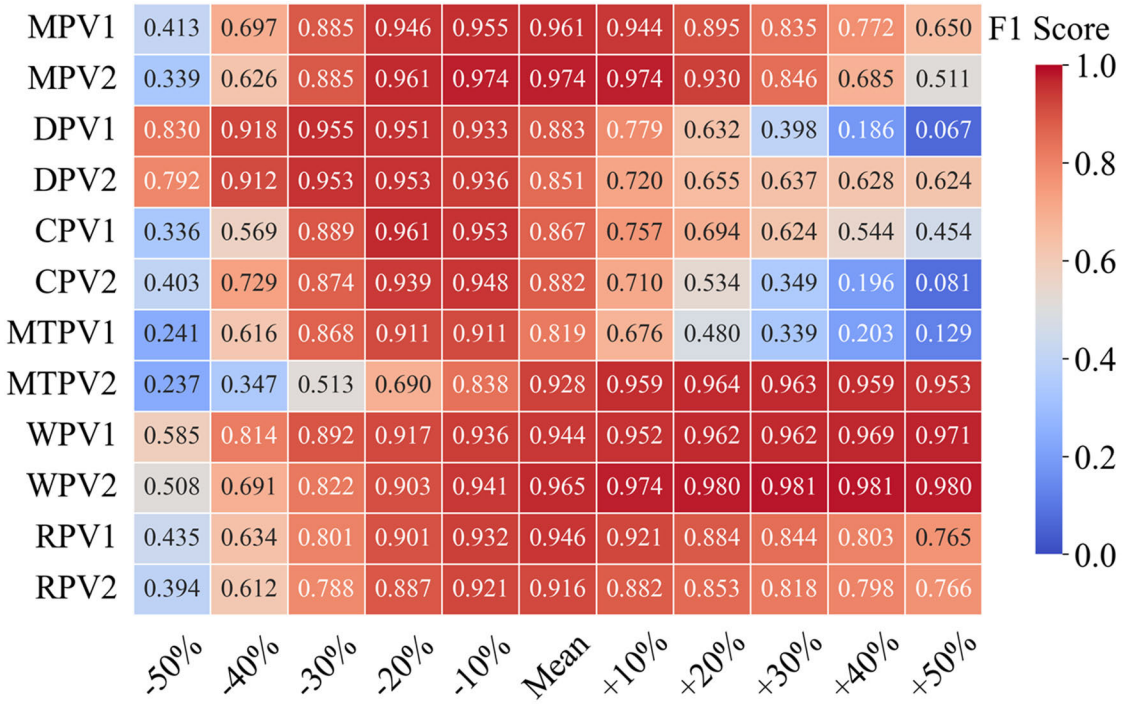


Fig. 14. Threshold sensitivity of APVI under different deployment scenarios. "Mean" refers to the average optimal threshold of APVI across the 12 cases shown in Fig. S16, which is 0.178. "±n%" indicates a threshold equal to the mean threshold ±n% of its value.

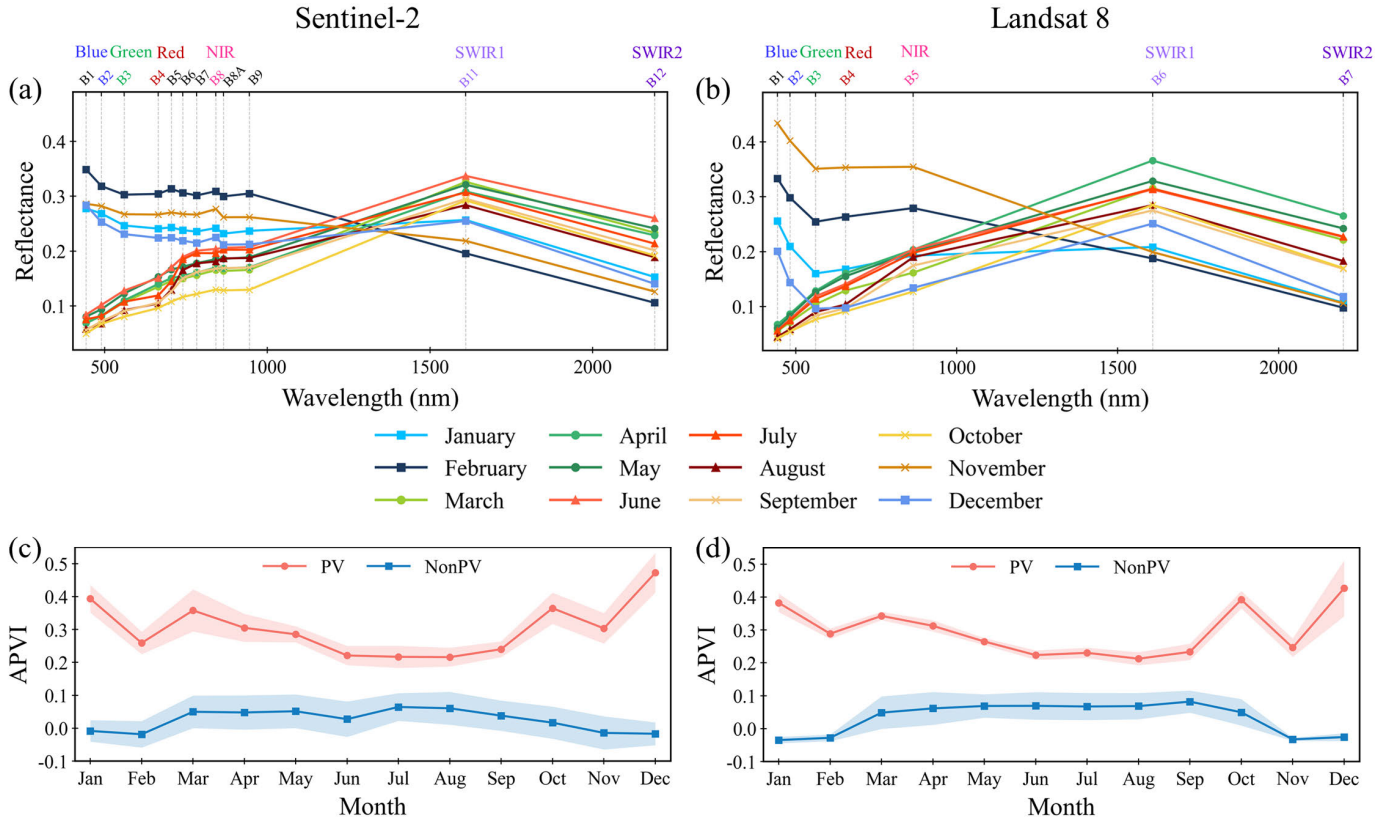


Fig. 15. Monthly variations of PV spectra (mean values) and APVI (mean and standard deviation values) based on (a) and (c) Sentinel-2 and (b) and (d) Landsat 8 imagery. Note: Landsat 8 bands B1–B7 (B1: coastal aerosol, B2: blue, B3: green, B4: red, B5: NIR, B6: SWIR1, and B7: SWIR2) correspond to Sentinel-2 bands B1–B4, B8, B11, and B12.

derived from both satellite datasets effectively identifies PV arrays across all seasons, indicating that the index is not only applicable to Landsat 8 but also demonstrates strong seasonal robustness. Moreover, PV arrays in the northern

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part of the spoil dumps are under continuous development, and APVI successfully tracks their deployment progress, further highlighting its potential for real-time PV monitoring. Nevertheless, the 30 m spatial resolution of Landsat 8 can introduce stronger mixed-pixel effects than Sentinel-2, especially along PV boundaries, small arrays, and fragmented patches. Therefore, although Landsat 8 and Sentinel-2 show generally consistent APVI responses and PV mapping patterns, differences in boundary details and small-patch detection may still occur because of their different spatial resolutions (see Fig. S25).

Building on this, we further tracked the monthly changes in both the spectral characteristics and the APVI of PV arrays (see Fig. S26). It is important to note that the sampled region was not a single pixel but a collection of pixels within a circular area with a diameter of 60 m centered on the sampling point. This approach provides a more comprehensive representation of the overall changes in PV arrays and their underlying surfaces, avoiding the randomness associated with individual pixels. As shown in Fig. 15, the spectral profiles of PV arrays in both satellite datasets exhibit similar variations across different months, demonstrating clear temporal heterogeneity. Specifically, from spring to summer (March–August), the underlying surface transitions from bare soil to vegetation, resulting in an increased spectral slope from the red to the NIR bands (as observed in August). From summer to snow-free autumn months (June–October), the underlying surface shifts back from vegetation to bare soil, leading to a gradual decrease in spectral slope. Therefore, phenological changes in the underlying vegetation significantly influence the spectral characteristics of PV arrays in the visible to NIR range.

During the snow-covered months (from November to February of the following year), the spectral characteristics of PV arrays are generally consistent with those shown in Fig. 3. Interestingly, we found that when PV arrays show high reflectance in the visible to NIR regions (e.g., in January), their NIR reflectance is higher than that in the SWIR1 band. Conversely, when reflectance in the visible to NIR regions is low (e.g., in December), the NIR reflectance becomes lower than that in the SWIR1. This phenomenon may be related to the degree of snow cover. Since snow typically has high reflectance, a higher snow cover fraction in PV pixels tends to make them exhibit more snow-like spectral features, thereby altering the spectral characteristics from NIR to SWIR1. Therefore, snow primarily affects the spectral response of PV arrays in the visible to SWIR1 range.

The monthly variation curve of APVI shows that the index, calculated from Sentinel-2 and Landsat 8 imagery, effectively distinguishes between PV arrays and the background in all months [see Fig. 15(c) and (d)]. Even during the vegetation growing season (June to September), although the APVI value slightly decreases due to the mixing of underlying surface vegetation pixels, it still maintains good discriminatory ability. These results indicate that, in medium-resolution satellite imagery, PV arrays exhibit not only spatial heterogeneity but also significant temporal variation. The proposed index effectively mitigates the spatiotemporal variability of the PV spectra, enabling robust extraction of PV arrays.

Although the proposed index is constructed based on three spectral features of medium-resolution PV pixels, these features are also evident in pure PV spectra, as illustrated by the hyperspectral reflectance curves of PV panels from He et al. [39] (see Fig. S1), particularly the flat spectral shape between the red (B4) and NIR (B8) bands. However, this feature is not always apparent in satellite-scale PV pixel spectra, especially for PV arrays deployed in vegetated areas. The mixing of vegetation pixels can dilute or distort the flat spectral shape feature of PV, making it difficult to detect or enhance this feature in medium-resolution imagery. In contrast, when the influence of underlying surface mixing is minimal (i.e., in pure PV pixels), the flat spectral feature becomes more prominent, and the term abs[(B8 − B4)/(B8 + B4)] in APVI can then effectively enhance this feature, resulting in higher index values for PV panels. Therefore, although APVI is designed based on coarse spatial resolution satellite imagery spectra, it still holds the potential to accurately identify and extract pure PV panels, provided the imagery has sufficiently high spatial resolution (e.g., finer than the width of the PV panels) and includes the necessary spectral bands.

C. Comparison With Existing PV Datasets

To more intuitively demonstrate the reliability of the proposed method, we further compare it with five existing PV datasets. These datasets were generated using deep learning or machine learning approaches. In the datasets by Li et al. [18] and Chen et al. [28], the most recent PV data are from 2022 and were derived from Sentinel-2 imagery. The datasets by Feng et al. [64] and Zhang et al. [27] contain PV data up to 2020, derived from Sentinel-2 and Landsat 8 imagery, respectively. In the dataset by Kruitwagen et al. [29], the latest PV data are from 2018 and were derived from SPOT imagery.

Due to the differences in the PV deployment periods covered by each dataset, Sentinel-2 imagery from the corresponding time periods was used in this study to support a more consistent visual comparison. Across different time points, the PV vectors obtained using APVI show good agreement with the actual PV distribution observed in Sentinel-2 imagery. In contrast, the existing datasets exhibit varying degrees of omission or overclassification in some complex scenarios. For example, Feng et al. [64] and Li et al. [18] miss certain PV arrays in mining areas, croplands, and deserts, as indicated by the green arrows in Fig. 16(a), (c), and (d). Similarly, Chen et al. [28] and Zhang et al. [27] show some overclassification of PV arrays on water surfaces and in desert regions, as indicated by the red arrows in Fig. 16(b) and (d).

It should be noted that these differences should not be interpreted as being caused solely by the deep learning or machine learning methods themselves. Existing PV datasets differ in terms of training samples, reference labels, input imagery, spatial resolution, temporal coverage, minimum mapping units, postprocessing strategies, and original mapping objectives. These dataset-related factors can also contribute to omissions, overclassification, or differences in boundary delineation. Therefore, the comparison in this section is intended as a dataset-level comparison rather than a strict method-to-method benchmark under identical experimental conditions.

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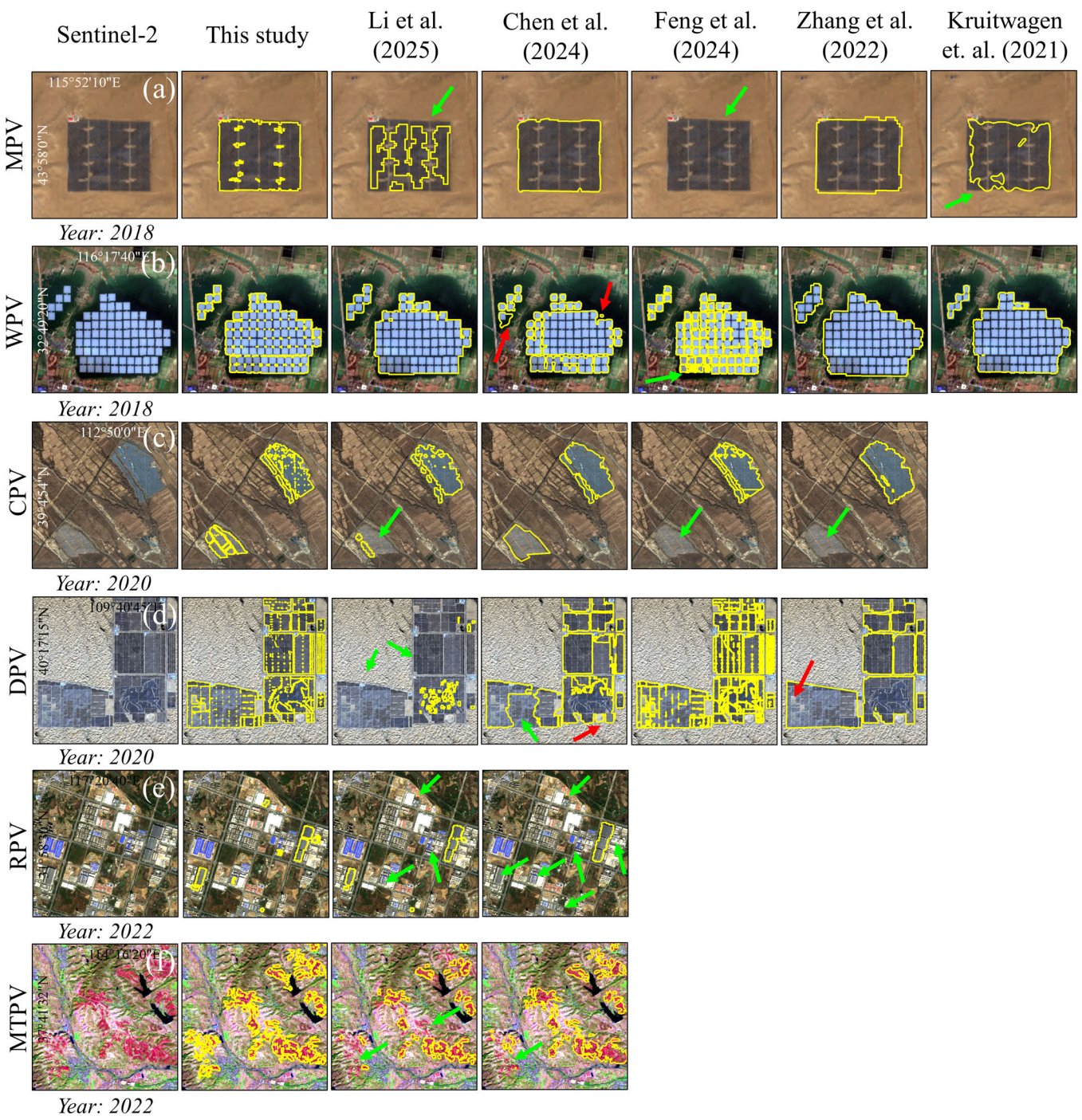


Fig. 16. Comparison between PV vectors extracted by the proposed method and those from existing datasets. Yellow vector boundaries indicate the mapped PV areas. Green arrows indicate omitted PV areas, and red arrows indicate overclassified PV areas. To clearly display the locations of PV installations, the basemaps in (a)–(e) are true-color Sentinel-2 images for the corresponding years (R:G:B = red:green:blue), while (f) uses a false-color Sentinel-2 image (R:G:B = SWIR1:NIR:Red).

The results generated in this study demonstrate a relatively high level of spatial detail. Specifically, thanks to the strong background suppression capability of APVI, the proposed results provide more refined delineation of PV array boundaries and internal structures than the existing datasets. For example, for PV extraction in mining areas, the PV patches generated by the proposed method exhibit a "hollow" structure consistent with the remote-sensing imagery, effectively suppressing the large gaps between PV panels. In contrast, the datasets produced by Chen et al. [28], Zhang et al. [27], and Kruitwagen et al. [29] tend to depict the overall contours of PV farms. Moreover, APVI also demonstrates advantages in enhancing PV signals under complex background conditions. In the urban rooftop area shown in Fig. 16(e), APVI enables more complete extraction of distributed PV installations, reducing omissions. In mountainous regions with complex

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terrain, APVI also provides more comprehensive mapping of irregularly distributed PV systems.

Overall, the proposed method can effectively handle complex PV deployment backgrounds, and the resulting PV vectors exhibit good regional completeness and spatial detail. However, this does not imply that APVI is intended to replace existing methods. Instead, APVI can potentially serve as an auxiliary spectral feature or prior information for machine-learning- and deep-learning-based methods, thereby improving PV mapping under complex backgrounds and diverse deployment conditions.

D. Application of APVI in Representative Regions of Other Countries

To further demonstrate the generalization capability of APVI, we conducted additional tests. As shown in Fig. S27, six representative PV power plants were selected from different continents: Nunez de Balboa solar farm in Spain (a), Jasper and Lesedi Solar Power Plant in South Africa (b), Taean Solar Power Plant in South Korea (c), urban rooftop PV systems in Melbourne, Australia (d), California Valley Solar Ranch in USA (e), and PV plants in the Atacama Desert, Chile (f). These external sites were not used in the development of APVI or in the derivation of the mean threshold used in the threshold-sensitivity analysis. Instead, they were used as independent test sites to evaluate the generalization capability of APVI under different geographic and environmental conditions.

For these external sites, the threshold was determined independently for each site using the same threshold-selection procedure described in Section IV-B, rather than directly applying the mean threshold derived from the Chinese study areas. As shown in Table S7, the optimal thresholds differ among the six external sites, reflecting the influence of regional differences in PV materials, background land covers, climate conditions, and image acquisition characteristics. Nevertheless, APVI successfully distinguishes PV arrays from the background in all external test sites, with F1 scores exceeding 0.908. These results indicate that APVI has good generalization potential beyond China, while local threshold fine-tuning remains necessary for obtaining optimal PV mapping results.

E. Justification of Class Separability and Optimal-Threshold Evaluation Metrics

To further examine the rationality of the class separability metric used in this study, we compared the JMD with the separability index (SI). JMD was selected as the primary separability metric because it is widely used in remote-sensing studies and has a bounded range from 0 to 2, which facilitates consistent interpretation across different classes and indices. In general, a larger JMD value indicates better class separability, and values greater than 1.8 are commonly regarded as indicating strong separability.

In addition, SI was introduced as a supplementary metric to further validate the separability results. SI measures the ratio between inter-class difference and intraclass variability, and a larger SI value indicates better separability [65]. The SI is calculated as follows:

$$\mathrm{SI} = \frac{\Delta\mathrm{inter}\,(i,j)}{\Delta\mathrm{intra}\,(i,j)} = \frac{|\overline{\mu}_i - \overline{\mu}_j|}{1.96 * \left(\sigma_i + \sigma_j\right)} \tag{13}$$

where SI represents the separability value between class i and class j. $\overline{\mu}_i$ and $\overline{\mu}_j$ represent the average of class i and class j, respectively. σ_i and σ_j represent the standard deviation of class i and class j.

As shown in Figs. S28 and S29, the JMD- and SI-based results show generally consistent patterns. APVI achieves JMD values greater than 1.797 for all PV-background pairs, indicating strong separability. The SI results further confirm this pattern, with APVI achieving SI values greater than 1.091 across all scenarios. By comparison, EPVI also performs well in many cases, but shows weaker separability in some WPV and snow-covered scenarios. These results indicate that APVI provides robust separability between PV and background classes under diverse deployment conditions. Nevertheless, because SI has no fixed upper bound and its numerical range varies more substantially among class pairs, JMD was retained as the main separability metric, while SI was used as a supplementary confirmation.

In addition to the F1 score, we further tested the Matthews correlation coefficient (MCC) and the area under the precision–recall curve (AUC-PR) to examine the rationality of the optimal-threshold selection. MCC provides a more comprehensive evaluation under class-imbalanced conditions because it considers all four elements of the confusion matrix, including TPs, TNs, FPs, and FNs. The MCC is calculated as follows:

$$\mathrm{MCC} = \frac{\mathrm{TP} \times \mathrm{TN} - \mathrm{FP} \times \mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})\,(\mathrm{TP}+\mathrm{FN})\,(\mathrm{TN}+\mathrm{FP})\,(\mathrm{TN}+\mathrm{FN})}} \tag{14}$$

where TPs, FNs, FPs, and TNs refer to the number of PV pixels correctly detected, PV pixels omitted, non-PV pixels wrongly identified as PV, and non-PV pixels correctly excluded, respectively, based on the comparison between classification results and reference points.

AUC-PR was also used as a threshold-independent evaluation metric by measuring the area under the precision–recall curve. This metric is particularly useful for evaluating PV extraction performance under imbalanced PV and non-PV samples.

As shown in Fig. S30, the F1 and MCC curves show highly consistent threshold-response patterns across different indices and deployment scenarios. Table S8 further shows that the optimal thresholds derived from F1 score and MCC are consistent across all tested scenarios. This indicates that the threshold selection is robust and not dependent only on the F1 score. In addition, APVI achieves the highest AUC-PR values under all tested scenarios, including MPV1, DPV1, CPV1, MTPV1, WPV1, RPV1, MPV1_winter, and CPV1_winter. These results demonstrate that APVI maintains superior PV extraction performance not only at the optimal threshold but also across the full threshold range.

Overall, the additional MCC and AUC-PR analyses further confirm that APVI outperforms the existing indices in both threshold-dependent and threshold-independent evaluations.

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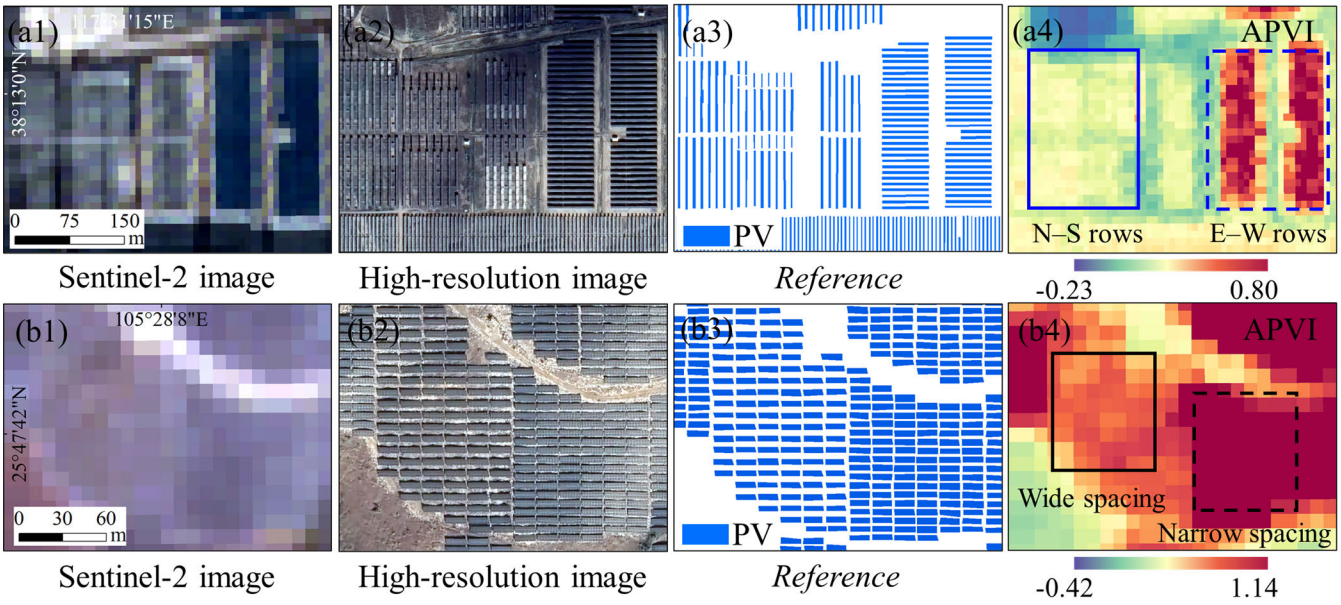


Fig. 17. Effects of PV panel orientation and panel spacing on APVI. (a) Effect of PV panel orientation. (b) Effect of PV panel spacing. (a1) and (b1) Sentinel-2 true-color images, (a2) and (b2) high-resolution Google Earth images, (a3) and (b3) reference PV maps, and (a4) and (b4) APVI images. In (a4), the blue solid rectangle indicates PV arrays arranged in north–south (N–S) rows, whereas the blue dashed rectangle indicates south-facing PV panels arranged in east–west (E–W) rows. In (b4), the black solid rectangle indicates PV arrays with wide panel spacing, whereas the black dashed rectangle indicates PV arrays with relatively narrow panel spacing.

F. Uncertainties

Although the proposed APVI has achieved satisfactory results, several factors still affect its application. First, the layout of the PV panels itself is a potential factor. Most PV panels are installed facing south to maximize solar radiation absorption. However, some PV panels are oriented east-west. As shown in Fig. 17(a3), PV panels facing south have higher APVI values than those facing east–west. This is because the Sentinel-2 satellite is in a sun-synchronous orbit, and the image contains a higher proportion of PV panels oriented south. Furthermore, the tilt angle of PV panels (i.e., the angle between the panel and the horizontal surface) is adjusted according to terrain, latitude, and season to better align perpendicular to the direct solar radiation, thereby receiving more radiation and sunlight duration. Typically, in high-latitude regions, PV panels have a higher tilt angle in winter [66]. At the same time, an increase in the tilt angle of the panels leads to a larger shadow between the panels, so the spacing between the panels needs to be increased to avoid shadow overlap [67]. As shown in Fig. 17(b3), the PV panels on the left have a larger tilt angle and greater spacing than the panels on the right. This results in more significant influence from underlying surface pixels in the Sentinel-2 images, which in turn lowers the APVI values.

Although the proposed method demonstrates robust PV identification across various deployment scenarios and under snow-covered conditions, the applicability of the APVI threshold is still affected by deployment context. Applying a fixed threshold across all scenarios may not yield optimal extraction results. However, as discussed in Section VI-A, APVI exhibits good threshold stability. In cases where the deployment scenario is unknown, a relatively relaxed threshold can be used for initial PV detection. Subsequently, the threshold can be fine-tuned according to local background conditions, PV panel characteristics, and seasonal variations to achieve more accurate mapping results. In addition, incorporating temporal information, such as seasonal patterns, can further help reduce misclassification.

Finally, this study detects PV installations based on medium-resolution satellite imagery, which captures not only the PV panels but also parts of the underlying surface, possibly leading to a slight overestimation of PV area. This issue is mainly due to the coarse spatial resolution of Sentinel-2 imagery. As discussed in Section VI-B, APVI shows strong potential in identifying pure PV panels. With the advancement of satellite imaging technologies, it will be possible to accurately extract PV panels once multispectral imagery achieves high-enough spatial resolution. In addition, a practical and efficient strategy is to first perform rapid detection of PV arrays using APVI based on medium-resolution imagery, and then to refine the mapping of PV panels using high-resolution imagery (e.g., Google Earth images).

This study aims to investigate the spatiotemporal spectral characteristics of PV arrays in medium-resolution remote sensing imagery and to provide a rapid detection method suitable for complex backgrounds. Although the proposed method has certain limitations, it can still serve as an important complement to existing PV remote sensing techniques, benefiting from the spatiotemporal continuity of satellite imagery and the capabilities of the GEE platform.

VII. Conclusion

This study proposes a novel PV index, the APVI, developed through a systematic analysis of the spectral characteristics of PV under various deployment scenarios and snow-covered conditions. In medium-resolution Sentinel-2 satellite imagery,

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PV exhibits pronounced spatiotemporal heterogeneity due to different interpanel land cover types and temporal variations. The comprehensive spectral analysis reveals that various types of PV exhibit consistent spectral characteristics in the SWIR1-to-SWIR2 bands. The red, NIR, SWIR1, and SWIR2 bands are the sensitive bands for distinguishing between PV and background land cover types. The results of visual evaluation, separability analysis, and PV mapping, compared with the existing five indices, show that APVI exhibits excellent performance superior to the existing indices across various scenarios. APVI can highlight PV in different scenarios while suppressing various background land cover types, achieving high separability between PV and the background land cover types. APVI not only demonstrates high mapping accuracy across various PV deployment scenarios (with an F1 score of 95.2%–95.2%), but it is also the only index capable of accurately identifying PV in snow-covered scenarios. In snow-covered scenarios, the accuracy is improved by 29.3–34.6 percentage points compared to existing methods. These results indicate that the index can effectively handle the uncertainty caused by the temporal and spatial variability of the underlying surface. Threshold sensitivity analysis shows that APVI can maintain relatively stable thresholds across different scenarios. Due to the use of common spectral bands, APVI can be effectively applied to other medium-resolution satellite imagery (e.g., Landsat imagery). The PV mapping accuracy of over 90% in typical power plants of different countries demonstrates that APVI holds great potential for cross-regional PV mapping. Compared with existing datasets, APVI shows distinct advantages in delineating the structural details of PV arrays and extracting small-scale distributed PV installations. In conclusion, APVI provides an accurate, robust, and efficient solution for detecting PV arrays across complex deployment scenarios and temporal variations.

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